Professional Networks and the Labour Market Assimilation of Immigrants*

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Abstract

This paper examines how professional networks shape immigrants' job search and labour market assimilation. Analysing displaced workers in Sweden, we show that connections to firms through conationals are particularly valuable for immigrants' job finding. As networks expand over time to include more natives, the value of such connections increases to that seen for displaced native workers. Origin homophily is also an important predictor for whether a connection provides a referral, associated with higher earnings and lower turnover, as opposed to simply providing information about vacancies. Networks account for around 20 per cent of the growth in re-employment rates among immigrants over their first ten years after arrival. After 15 years the productivity of immigrants' professional networks is comparable to those of natives.

Keywords: immigrant labour market integration, professional networks, job search
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1 Introduction

Immigrants in advanced economies face persistent employment and earnings gaps relative to native-born workers. These disparities affect individual welfare, public finances, and the overall economic impact of immigration. Although the gaps narrow with time spent in the host country, substantial differences typically remain.[1] The literature suggests many potential barriers facing newcomers after migration, including lack of relevant language skills (Auer, 2018; Foged et al., 2024; Chiswick and Miller, 2015), non-transferability of foreign credentials (Brücker et al., 2021), or skill mismatch to local labour market demands (Eckstein and Weiss, 2004).

Access to productive social networks, in particular those formed in professional contexts, is a critical resource for overcoming such barriers. Networks serve two main purposes in the labour market: transmitting information about vacancies and job requirements (Montgomery, 1992; Granovetter, 1995; Calvó-Armengol and Jackson, 2004; Caldwell and Harmon, 2019), and providing referrals that reduce matching uncertainty for both workers and employers (Montgomery, 1991; Burks et al., 2015; Dustmann et al., 2016a). For migrants, such networks can be especially valuable when host-country-specific knowledge is limited, supplying insights into workplace norms, industry practices, and employer behaviour. As discussed below, previous work emphasises the role of co-ethnic connections for early outcomes. But over time, as immigrants accumulate local experience, networks may expand and diversify to include more natives or other migrant groups.[2] The relative importance and productivity of different ties are therefore likely to change as immigrants gain familiarity with the host country.

Drawing on comprehensive Swedish administrative data, we study how immigrants' professional networks evolve with time after arrival, and how this affects their job-search outcomes. We focus on workers displaced by a plant closure and use a "dyadic" empirical design to estimate the causal impact of different types of professional connections on re-employment and subsequent outcomes. We then use our estimates to quantify how changes in networks and connection effectiveness contribute to immigrant-native convergence in employment and resilience to economic shocks.

Our data cover all individuals aged 16–64 in Sweden during 1985–2018. Using individual and workplace identifiers, we link workers to their current and past employers as well as to former coworkers. We define a worker as connected to a firm if that firm employs someone with whom the worker overlapped in a previous workplace during the preceding five years. Our network measure thus captures job-related contacts accumulated through host-country work experience, rather than active networking effort. Consistent with immigrants' gradual labour market integration, we find that these coworker networks expand with time spent in Sweden. Their composition also

[1] Since Chiswick (1978), a large literature has consistently documented substantial but incomplete convergence to the outcomes of comparable natives with time spent in the host country, across national contexts and time periods. A non-exhaustive list of contributions includes Borjas (1985, 1995), Eckstein and Weiss (2004), Card (2005), Lubotsky (2007), Algan et al. (2010), Dustmann and Görlach (2015), and Rho and Sanders (2021).

[2] This reflects the distinction between "bonding" social capital—ties within one's origin group—and "bridging" social capital—ties that connect different groups, particularly to natives (Putnam, 2000).

changes: connections through conationals and other immigrants become less dominant, while the share of connections through natives rises steadily. The number of firms to which immigrants are linked through former native coworkers comes close to the corresponding number for natives. At the same time, immigrants continue to have more firm connections through foreign-born workers, pointing to persistent workplace segregation.

To estimate the causal effect of immigrants' networks, we implement a dyadic strategy that compares workers displaced by the *same* closure and tests whether a worker is more likely to be hired by a specific firm if they have a connection there (via former coworkers), relative to co-displaced peers lacking such links. By conditioning on closing-plant × potential-destination fixed effects and rich worker–firm covariates, we absorb systematic sorting between specific plants and firms, and isolate the causal effect of different connection types on hiring (cf. Eliason et al. 2023).[3] Focusing on workers displaced by plant closures mitigates concerns about endogenous job search: closures generate a plausibly exogenous transition into unemployment. Displaced immigrants in our data resemble the broader immigrant population in many dimensions, supporting external validity. Unemployed workers also rely more on networks than on-the-job searchers (Caldwell and Harmon, 2019), which heightens the policy relevance of our results.

In line with previous research, we find that workers are substantially more likely to join a firm where they have a connection, compared to non-connected workers affected by the same displacement. A clear and consistent pattern in our results is that the value of a connection is strongly related to the worker's country of origin. Among displaced natives, the effect of native connections on the hiring probability is roughly 50 per cent as large as that of immigrant connections. For immigrants, the impact of connections from their own origin group is roughly eight times larger than that of native connections, and also considerably larger than the effect of other migrant connections. Our analysis of other dimensions of homophily (gender, age, education) and other aspects of tie quality (recency, length, seniority, workplace size etc.) yields results in line with theoretical expectations, while at the same time consistently showing the excess importance of origin similarity. However, the effect of native connections grows over time, doubling during the first twenty years an immigrant spends in Sweden.

We then examine how earnings and subsequent turnover differ between connected and unconnected hires. Consistent with referral-based job-search models (Dustmann et al., 2016a; Glitz and Vejlin, 2021), displaced workers hired at firms where they have connections tend to earn more than unconnected hires at least in the short run, and to have lower job separation rates. There is, however, strong homophily also in these associations. Displaced natives hired through native connections have higher earnings, but this is not the case for those hired through immigrant connections. For immigrants, point estimates for conational and other migrant connections are substantial, but there is no sign that native connections provide referrals increasing entry wages.

[3]For other studies using plant closures to study professional networks, see, e.g., Cingano and Rosolia (2012) and Glitz (2017). Other work using dyadic designs includes (Kramarz and Skans, 2014) on youth labour-market entry and (Hafner and Hedtrich, 2023) on early scientific careers.

Finally, we quantify the total contribution of changes in the size, composition and effectiveness of professional networks to employment outcomes over time. Immigrants who have been in Sweden for 20 years are about six percentage points more likely to be employed in the year after the plant closure than those displaced in their first years in Sweden. We find that immigrants' connections raise their total re-employment probability by at least 1.0–1.5 percentage points after 20 years in Sweden. These figures correspond to roughly 17–25 per cent of the long-term convergence in the immigrant-native gap in the re-hiring probability and are comparable in magnitude to the total effect of natives' networks. More inclusive definitions of immigrants' professional networks imply larger total effects on re-employment, up to 5–6 percentage points. Our results therefore suggest that the accumulation of professional networks is an important driver of immigrants' labour market integration. Focusing on the total contribution of different types of connections, we find that while the number of native connections grows rapidly, the total productivity of conational contacts typically exceeds that of native ones in the early years in the host country.

Our paper contributes to and bridges two strands of research: (i) the labour market impact of ethnic and general networks; and (ii) the economic assimilation of immigrants and its underlying mechanisms. Numerous studies have documented the importance of initial networks—those present at the time of migration—for immigrants' successful labour market integration (Battisti et al., 2022; Beaman, 2012; Damm, 2009; Edin et al., 2003; Munshi, 2003). We extend this literature in three ways. First, we focus on individuals with whom immigrants are known to have interacted—former coworkers—and study how origin similarity and dissimilarity affect the value of a connection. Second, rather than treating networks as fixed at arrival and examining their long-term effects on integration, we analyse how changes in networks over time improve immigrants' labour-market outcomes.[4] Finally, these papers almost all focus on refugees, a group with very weak labour market attachment; we show that even immigrants with stronger attachment to the labour market benefit from having network connections.

Regarding the second strand of literature, the initial earnings gap between immigrants and native-born workers is often attributed to the depreciation of skills acquired in the source country (Dustmann et al., 2013, 2016b). The accumulation of host-country-specific human capital—most notably language skills (see, e.g., Foged et al., 2024; Auer, 2018; Chiswick and Miller, 2015)—contributes to the convergence of immigrant and native wages. Increasing returns to pre-existing skills provide a second mechanism (Eckstein and Weiss, 2004).[5] We contribute by showing that professional networks constitute an additional, complementary mechanism: they reflect the build-up of host-country-specific capital and exhibit changing returns over time, contributing to immigrants' employment and earnings growth. This mechanism may also help

[4]The only partial exception we are familiar with is Glitz (2017) who allows the effect of the employment rate of past coworkers to differ by the ethnicity of coworkers, but does not differentiate between recent and established immigrants.

[5]The various causes of the earnings gap between immigrants and native-born workers are not mutually exclusive. For instance, the recognition of foreign credentials significantly contributes to earnings convergence (Brücker et al., 2021) and likely works in tandem with the expansion of social networks.

explain recent findings on firm sorting and mobility—immigrants initially sort into lower-wage firms and gradually climb the job ladder, closing part of the gap (Arellano-Bover and San, 2023; Amior and Stuhler, 2023; Dostie et al., 2020; Åslund et al., 2025), potentially as a result of acquiring better networks through employment (Willis, 2025). Our analysis of network impacts following displacement also contributes to the growing literature on differential resilience to economic shocks among immigrants and natives.[6]

The paper is structured as follows. Section 2 outlines the data and central definitions. In Section 3 we describe the build-up of coworker networks and present summary statistics on displaced workers and the relationship between network size and re-employment post-displacement. Section 4 presents the dyadic framework and its estimates on the causal effect of connections on re-employment among displaced workers. The section also presents variations and robustness checks (including a placebo analysis), and investigates how connections are associated with post-employment outcomes. Finally, in Section 5 we analyse how changes in networks and the productivity of connections relate labour market assimilation in terms of immigrant-native convergence in re-employment and unemployment probabilities over years in Sweden. We conclude in Section 6.

2 Data

2.1 Data sources

The analysis is based on administrative data for the entire Swedish population 16–64 years old during 1985–2018. Data from multiple registries held by Statistics Sweden are linked using pseudonymised personal identifiers.[7]

A compilation of labour market statistics (RAMS, *Registerbaserad arbetsmarknadsstatistik*) contains core information and links employees to firms and workplaces (SCB, 2026d). These data include an employment register containing statutory income statements filed annually by each employer (firm and establishment) to the Swedish Tax Agency for each worker. We define employment as having positive annual earnings. The employment register also includes job-spell information, which we use to verify that displaced workers and their connections from previous workplaces (or the closing establishment) had overlapping spells. We discuss how coworker networks are defined in more detail below. Private-sector firms and establishments are identified using the firm register.

[6]Studies of job displacement typically find that immigrants experience larger post-displacement earnings or employment losses than natives (Hardoy and Schøne, 2014; Bratsberg et al., 2018; Illing and Koch, 2021), although Balgova and Illing (2023) show that controlling for pre-displacement outcomes can attenuate or alter these differences. They also provide suggestive evidence that differences in social networks may contribute to weaker post-displacement recovery among immigrants. More broadly, Berbée and Stuhler (2025) document immigrant employment fragility following structural labour-market shocks in Germany.

[7]The data sources are documented in Statistics Sweden's microdata documentation portal; see SCB (2026c).

We also use the Longitudinal integrated database for health insurance and labour market studies (LISA), from which we obtain gender, birth year, and education for the entire population (SCB, 2026b).[8] County of birth for Sweden-born individuals and country-of-birth groups for foreign-born individuals come from a table based on the Total Population Register (SCB, 2026a).[9] We use immigration and emigration dates from RTB to define the year of first immigration for the foreign-born population.

2.2 Sample and definitions

2.2.1 Annual employment panel

RAMS provides monthly employment-spell data. We construct an annual panel based on each individual's main job in the year. The main job is the November workplace (the reference month for official statistics) that yields the individual's highest annual earnings. If an individual is not employed in November, no main job is defined. Subsequent analyses use this panel to construct firm characteristics and individual networks, and to identify closures.

2.2.2 Identification of closing workplaces

We use the main-job data set to identify workplace closures. We define a closure between November of year t and November of year $t+1$ if a workplace employs some individuals in November of t but not in November of $t+1$ or $t+2$. To ensure that we do not misclassify either mergers or changes of workplace identifier (e.g. due to changes of ownership) as closures, we require that no workplace in $t+1$ employs more than 30 per cent of the workforce in year t of the closing firm (following Hethey-Maier and Schmieder, 2013). This definition of a closure, combined with the other data requirements for our analysis, implies that we study closures of firms whose last year of operation is 1990–2016.

To further ensure that closures are well-defined, we apply the following restrictions. We limit our attention to firms (i) that were employing at least 4 workers in the final year of operation, year t; (ii) composed of a single establishment; (iii) where firm and plant identifiers are non-missing; and (iv) operating in the private sector. We also drop firms that are defined as closing more than once in the data.[10]

Our main sample includes 50,212 closure events satisfying the above restrictions. Table A.1 in the appendix presents some statistics on the firms. The average firm had 10.3 employees in

[8] For the years 1985–1989, we collect the corresponding information from SYS, an IFAU-specific register compiled by Statistics Sweden from RAMS, the Education Register (UREG), the Total Population Register (RTB) and the Income and Taxation Register (IoT). SYS is not separately documented in Statistics Sweden's public register documentation.

[9] For confidentiality, birth-country information is aggregated into 28 categories; major source countries, such as Chile, Finland, Iran, Iraq, and Turkey, typically form their own category.

[10] We use the terms firm, workplace and establishment somewhat interchangeably. To be precise, all networks are defined at the workplace (i.e. establishment or plant) level, using the workplace identifier in the RAMS data, as are connections and hires. When studying closures, we limit ourselves to closures of single-workplace firms, making the distinction superfluous.

its final year of operation. Almost 60 per cent of the closures take place in the retail, hospitality and corporate services sectors. The table also reveals that a substantial share of the closing firms only had native or only immigrant employees, even though the segregated firms are on average much smaller than the mixed firms. The immigrant-only firms are often found in hospitality and exhibit a more polarised educational composition of the workers.

2.2.3 Definition of coworker networks

We measure professional connections using former coworkers from prior workplaces who are currently employed at other firms. In the context of plant closures, such connections may provide displaced workers with information or referrals that improve re-employment prospects.

In each year t, we identify an individual's past workplaces in the previous five years, limiting attention to workplace-years in which the individual had no more than 100 coworkers. Both conceptual and practical considerations motivate these limits. Interactions in the distant past are less likely to remain relevant for job search, and workers in very large workplaces are unlikely to interact intensively with all coworkers (see, e.g., Eliason et al., 2023). Including such workplaces would therefore both weaken the interpretation of observed coworker links as meaningful contacts. With the dyadic setting, it would also impose prohibitive computational costs. We nevertheless show below that this restriction does not differentially affect immigrants' and natives' observed sets of past coworkers, and that our main results are robust to more inclusive definitions of past coworkers. The size cap applies only to connection-forming workplaces; closing firms and potential destination employers may be larger.

Next, for each individual in year t (pre-displacement), we identify all workplaces that currently employ at least one of that individual's former coworkers. We exclude from this set the individual's own closing firm and the last workplace where that coworker and the individual previously worked together. Below, "connections" refer to links to firms via past coworkers, not to links between two individuals. We characterise each firm-level connection using the attributes of the coworker(s) who create the link.

Our definition of a connection captures potential job contacts created through past employment relationships, rather than deliberate networking effort. Measured network size therefore reflects prior employment in Sweden, job mobility across workplaces, workplace size, and the extent to which former coworkers disperse across firms. Our definition is deliberately more restrictive than is common in the professional networks literature (e.g., Cingano and Rosolia, 2012; Caldwell and Harmon, 2019; Glitz and Vejlin, 2021). In particular, we exclude coworkers from the closing workplace who moved to another establishment prior to the closure. Given our empirical strategy, such links risk confounding the estimated effects with shared firm-specific human capital among all displaced workers.

2.2.4 Analysis sample

Our main analysis includes all individuals displaced by a plant closure who satisfy the following restrictions: (i) aged 21–64 at the time of displacement; (ii) either born in Sweden, or immigrated after age 16 and have been in Sweden for up to 25 years; (iii) have not been employed at firms lacking well-defined workplace identifiers in the past five years;[11] (iv) have complete information on all control variables used in the main analysis. The final sample comprises 323,684 displaced individuals (Table 1).

By restricting our focus to workers displaced by plant closures, we follow a common strategy in the literature on the causal effects of professional networks (e.g. Cingano and Rosolia, 2012; Glitz, 2017). This restriction ensures that job search begins exogenously at the time of displacement, mitigating selection concerns that arise for voluntary job changers, who may be positively selected on the quality of their networks.

3 Descriptive results

3.1 Networks and employment in the population

Figure 1 uses data for the Swedish working-age population in 1990–2016 to describe basic patterns motivating our study: immigrants' gradual entry into employment and accumulation of a set of recent coworkers. In each observation year, we weight the native population according to the age distribution in each immigrant cohort, to create a cohort-specific comparison group. We then pool observation years in each figure and report outcomes for immigrants by years since migration and for comparable, age-weighted natives.

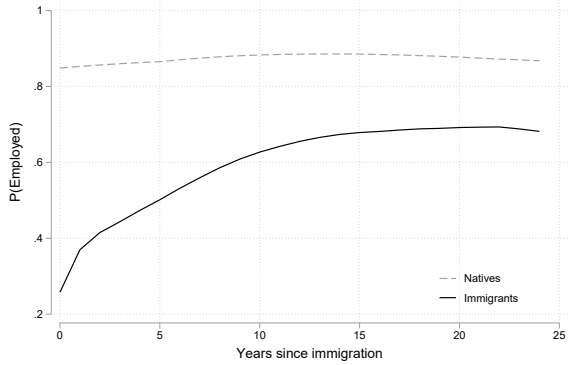
In Figure 1a, we document that the immigrant-native employment gap narrows from 59 percentage points in the year of immigration to 19 percentage points after 20 years in Sweden; i.e., the substantial but incomplete convergence typically seen in other contexts. Entry into the labour market also creates workplace-based contacts that may later serve as potential job connections. Our network measure is constructed from former coworkers observed over the previous five years; the number of such contacts therefore rises with immigrants' entry into and progression through the labour market. To illustrate how this process unfolds over time, we calculate the median number of former coworkers over the past five years, including individuals who were never employed or who have no former coworkers under this definition. Figure 1b shows that the number of recent coworkers among immigrants increases rapidly during the first few years in Sweden. After 9 years, immigrants catch up to same-aged natives in their median

[11] Workplace identifiers are not defined for firms in certain sectors, e.g. public sector employers or construction firms. The share of individuals having recently worked at such a firm in the previous five years falls over time, from around 24 per cent in 1990 to around 12 per cent in 2016, and is generally around 2 percentage points higher for natives. Our estimates are not sensitive to instead including these individuals in the sample and recording them as having no coworkers in that workplace-year (see Table A.4, column 3).

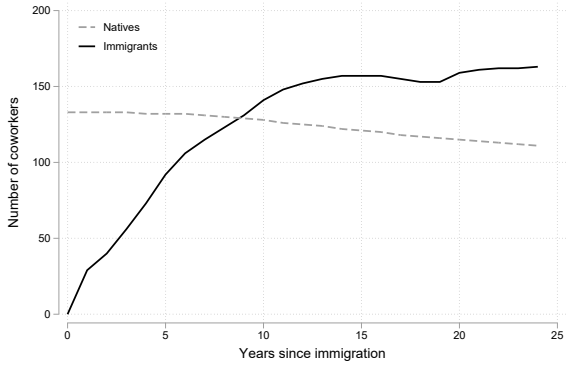
total number of former coworkers, although immigrants constitute a larger share of immigrants' former coworkers than of natives', as seen in Figure 1c.[12]

Figure 1: Number of recent coworkers and employment by years since immigration

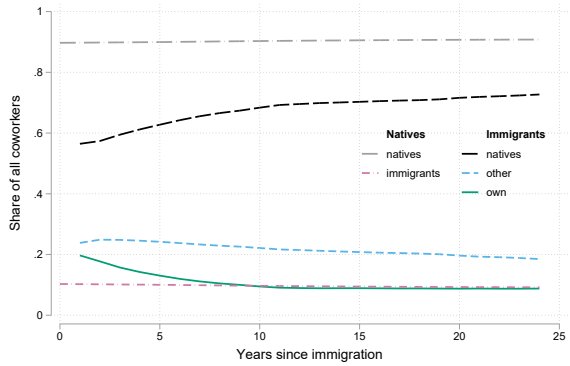
(a) Employment rate



(b) Number of recent coworkers



(c) Share of coworkers by type



Notes: Employment histories for natives and immigrants aged 16 or older at arrival for y in 1990–2016. All individuals are aged 21–64 in the observation year and are not missing a workplace identifier in the past five years. Figure 1b shows the median number of distinct coworkers during $[y-5, y-1]$, including individuals with no recent coworkers. Figure 1c reports the average share of recent coworkers of different types for immigrants and natives. "Others" refers to non-conational immigrants. To account for the correlation between age and years since migration, values for natives are weighted by the age distribution of immigrants observed at different years since migration.

The evolution of our network measure is directly linked to individuals' employment histories in Sweden. In particular, the number of coworkers accumulated over the five-year reference window depends on whether individuals are employed, how frequently they change workplaces, and the size of the workplaces where they work. To illustrate these determinants of the number of *recent past coworkers*, Figure A.1 reports how employment rates, workplace transitions, tenure, and workplace size vary with years since migration. On average, immigrants have a lower employment rate over the five-year reference period than same-aged natives, even after fifteen or more years in Sweden. Differences in turnover, conditional on past employment, are smaller. In the first five years in Sweden, immigrants move into new firms somewhat more often than natives, but these differences are small and do not persist. The continued increase in the number of immigrants' past coworkers after ten years in Sweden, relative to same-aged natives, is instead driven mainly by growth in workplace size over the five-year reference period.

[12]The negative slope for same-aged natives may seem counter-intuitive if one expects networks to grow over the life cycle. With a focus on recent coworkers, however, it is reasonable: younger individuals are more likely to work in larger firms and to change jobs more frequently.

3.2 Displaced workers

3.2.1 Summary statistics

Table 1 presents our main sample of displaced workers.[13] We differentiate workers by place of birth (Sweden or elsewhere) and, for immigrants, by years since immigration. Immigrants who experience a displacement in their first 5 years in Sweden are on average 4.8 years younger and 5 percentage points less likely to be females than displaced natives. These differences are consistent with established differences between immigrants and natives in female labour force participation and with the fact that immigrants tend to arrive while young. Age at displacement varies substantially with years since migration (as expected); in descriptive figures we therefore always present age-adjusted comparisons between immigrants and natives, using the weighting procedure described above. The differences in educational attainment are also substantial. The educational distribution of recent immigrants is more polarised, as immigrants are both more likely to have compulsory education and more likely to have tertiary education than natives. These differences are, however, less pronounced for immigrants with longer periods of residence in Sweden.

Natives have a stronger attachment to the labour market and their firm at the time of displacement: on average, natives have worked at 1.9 establishments in the five years before the displacement, have been employed for 4.2 out of those five years and have been employed at the closing firm for on average 1.7 years. For immigrants, the corresponding numbers are almost always lower, but differences to natives decrease with years since migration. The data also reveal that about 4 per cent of the migrants displaced within the first 5 years in Sweden emigrate in the following year. This figure falls to around 1 per cent for immigrants displaced after longer in Sweden.

A potential concern with our approach is that displaced workers are a selected group forced to look for work, limiting the broader relevance of the findings. A first argument against this view would be that access to and use of networks among active job seekers is particularly relevant. Second, it seems that particularly among immigrants the displaced workers are quite similar to the overall immigrant population (see Table A.2). While men are more likely to work in the private sector (Table A.1) and are therefore overrepresented among the displaced, factors directly related to the labour market position (education, past employment experience, tenure) are typically of the same magnitude. The most notable difference between our displaced workers and the typical employed individual, irrespective of nativity, is that the displaced individuals have many fewer former coworkers (see also Figure A.2). This reflects that displaced individuals have typically worked at smaller workplaces in pre-displacement years than the average worker.[14]

[13] Workers subjected to more than one displacement are treated as independent observations. With the sample criteria used for the analysis, 6.4 per cent of the individuals are exposed to multiple closures.

[14] This is in turn because our definition of closure, which excludes downsizing and plant splitting, implies that very large workplaces, which employ a disproportionate share of the labour force, almost never close.

Table 1: Summary statistics for displaced workers in year t

		Years since immigration				
	Native	0–4	5–9	10–14	15–19	20–24
Female	0.36	0.31	0.33	0.33	0.30	0.29
	(0.48)	(0.46)	(0.47)	(0.47)	(0.46)	(0.45)
Age	37.6	32.8	35.7	39.6	43.3	47.2
	(12.2)	(8.35)	(8.25)	(7.42)	(6.75)	(5.86)
Compulsory education	0.22	0.41	0.31	0.31	0.32	0.34
	(0.42)	(0.49)	(0.46)	(0.46)	(0.47)	(0.48)
Secondary education	0.65	0.33	0.45	0.46	0.51	0.51
	(0.48)	(0.47)	(0.50)	(0.50)	(0.50)	(0.50)
Post-Secondary education	0.13	0.26	0.25	0.22	0.18	0.14
	(0.33)	(0.44)	(0.43)	(0.42)	(0.38)	(0.35)
Number of workplaces, $[t-5, t-1]$	1.97	0.77	1.71	1.80	1.79	1.78
	(1.02)	(0.78)	(1.05)	(1.03)	(1.01)	(0.99)
Years employed during $[t-5, t-1]$	4.20	1.05	3.06	3.57	3.82	4.03
	(1.28)	(1.13)	(1.70)	(1.61)	(1.54)	(1.45)
Tenure in firm in t	2.67	1.48	2.07	2.34	2.56	2.68
	(1.93)	(0.81)	(1.48)	(1.73)	(1.87)	(1.94)
Emigrated one year later	0.0022	0.041	0.014	0.011	0.012	0.0075
	(0.047)	(0.20)	(0.12)	(0.10)	(0.11)	(0.086)
N connected workplaces	11.9	2.68	9.43	9.83	8.97	8.89
	(17.7)	(7.57)	(15.8)	(15.8)	(15.1)	(15.0)
N	278,993	18,064	10,418	6,669	5,157	4,383

Note: Mean coefficients; standard deviations in parentheses. The network variables measure the number of either firms or industries a displaced worker is connected to through past coworkers. Sex, education categories, and emigration by the end of $t+1$ are measured as indicator variables. Compulsory education includes individuals with missing education information.

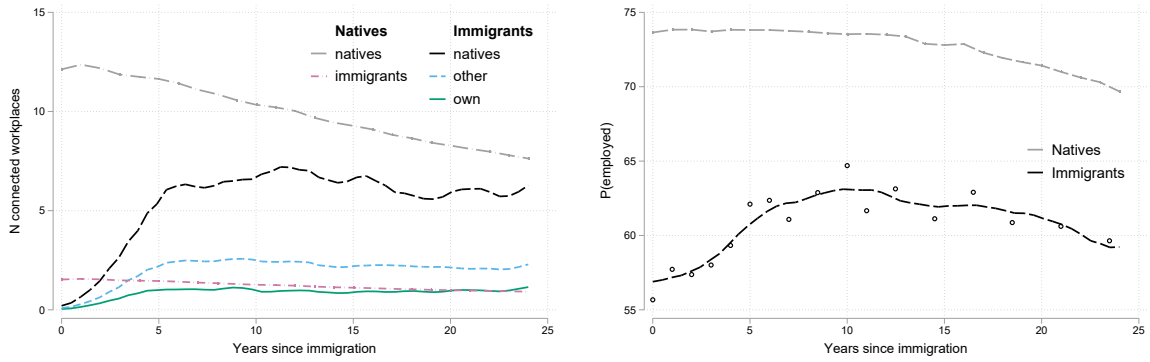
3.2.2 Networks and re-employment

Immigrant-native differences in recent labour force attachment translate to differences in the number of workplaces immigrant workers displaced at different durations of residence are connected to, shown in Figure 2a. The number of native connections rises rapidly and then levels off, implying convergence vis-à-vis same-aged displaced natives. The average number of immigrant connections—both conational and from other countries—plateaus after 5–10 years, and then declines slightly. Displaced immigrants' networks also become less segregated with time in Sweden; the share of native connections increases from 0.52 for immigrants displaced after a year in Sweden to around 0.62 for immigrants displaced after 15 or more years in Sweden (see Figure A.3).

We explore the determinants of the number of *connections* for displaced individuals in more detail in Table A.3. The table reports prior employment, past workplaces, total past coworkers, uncensored past coworkers, and the share of coworkers censored by the 100-coworker cap. Displaced immigrants are persistently slightly less likely to have been employed in recent years than natives, while the probability of having changed jobs at least once, conditional on prior employment, is similar after five years in Sweden. Very recent immigrants have many fewer past coworkers overall and are therefore less affected by censoring from large workplaces. For immigrants who have been in Sweden for at least five years, however, the extent of censoring is similar to that for natives. This suggests that the 100-coworker cap is unlikely to distort immigrant–native comparisons for the groups most relevant to the main analysis.

Figure 2: Connections and re-employment over time in host country

(a) Number of connections in t (b) Re-employment in $t+1$



Notes: Number of connections at displacement by years since immigration (Figure 2a) and re-employment probability of displaced workers as a function of years since immigration (Figure 2b). Kernel-weighted local averages are reported for immigrants, calculated using the Epanechnikov kernel and a default bandwidth. The average number of native connections, immigrant connections, and re-employment probability for natives are weighted by the age distribution of immigrants, grouping age in five-year bins, conditional on years since immigration. In Figure 2b we also report a binned scatter plot for immigrants, where ventiles of years since immigration are used as bins.

The re-employment probability (the share employed in November of $t+1$) increases from 57

per cent of those displaced between their first and second year in Sweden to around 60 per cent for those losing their job after five years in Sweden (Figure 2b). Thereafter, it plateaus at 63 per cent. Re-employment remains about 10 percentage points lower than among same-aged natives.[15] These substantial differences in job-finding are broadly consistent with the immigrant–native employment gap in the overall population; see Figure 1a. Figure 2 suggests that accumulating connections may contribute to improved post-displacement employment among immigrants. However, these associations are descriptive rather than causal. Indeed, the descriptive statistics in Table 1 show that recent and established immigrants differ along several observable dimensions, and established immigrants may also be more employable for unobserved reasons. We therefore turn to estimating the causal effects of different connection types on displaced immigrants' job-search outcomes.

4 Networks and job search outcomes of displaced workers

4.1 Empirical strategy

4.1.1 Presentation

To study how professional networks affect job search after displacement, we compare individuals affected by a plant closure and ask whether a firm is more likely to hire a displaced individual with a connection to the firm. For each closing firm j, we define a set of potential hiring firms k, which are all firms connected to firm j through a former coworker of a displaced individual. We then construct a set of dyadic observations for all workers displaced by the closure of j: one dyad for each displaced worker i and each potential hiring firm k. The outcome of interest is whether worker i is hired by firm k. The main explanatory variable is whether worker i has a connection to firm k. We define a connection as a past coworker of i being currently employed by k. We estimate the effect of such connections using the following linear probability model on the dyadic data:

$$Y_{ijk} = \sum_{h \in H} \sum_{n \in N} \gamma^{nh} c_{ik}^{n} c_{i}^{h} + \alpha_{jk} + X_{ik}'\beta + \epsilon_{ijk}. \tag{1}$$

Y_{ijk} is an indicator for displaced worker i, employed at firm j in year t, being employed at firm k in year $t+1$, multiplied by 100. The indicator c_{ik}^{n} allows the effect of a connection to differ by the type of past coworker creating the link, denoted by n, while c_{i}^{h} allows the effect to differ by type of the displaced worker, denoted by h. In the simplest case, one might distinguish only between displaced natives and immigrants and treat all connections as identical,

[15] Since our focus is on the role of connections (which by definition require potential coworkers in the new workplace) rather than on the labour market impact of displacement, we do not consider competing events, e.g. self-employment and transitions to studies or ALMPs.

i.e. $H = \{native, immigrant\}$ and $N = \{any\}$. In our main specification, we distinguish between native, conational, and other-immigrant connections. Our model also includes closing firm by hiring firm fixed effects, α_{jk}, and a vector of controls X_{ik}, which we discuss below.

The coefficients of interest in our model are the γ^{nh}. To interpret these coefficients, we adopt the constant-effects framework of Eliason et al. (2023). The idea is that, conditional on the worker–firm controls X_{ik} and the closing firm by hiring firm fixed effects α_{jk}, workers who differ in connectedness to firm k are assumed to have the same probability of being hired there in the absence of a connection. Under this identifying assumption, the estimated coefficients γ^{nh} capture the causal effect of a type-n connection to firm k on the probability of being hired there for a type-h worker. The coefficients should therefore be interpreted as the effect of having a connection to firm k relative to co-displaced workers who do not have that specific connection, rather than as the total effect of networks relative to a no-network counterfactual. We discuss the validity of our identifying assumption in the following subsection and return to the total effect of networks in Section 5.

All models we estimate include a fixed effect for the closing firm by hiring firm pair, α_{jk}. Because the model includes these fixed effects, the coefficients γ^{nh} are identified only from firm pairs where some displaced workers are connected to k and others are not. Firm pairs with no within-closing firm variation in connectedness are therefore not included in our estimation sample.[16] Intuitively, identification comes from comparing two workers displaced from the same firm j who face the same potential hiring firm k, but differ in whether they have a connection there. We also exclude cases where displaced workers are connected to k through more than one type of connection (e.g., both a native and an immigrant coworker). Having multiple types of connections to firm k suggests that some unobserved, type-invariant characteristic of the displaced worker's past coworkers may be causing clustered worker flows between firms, which would confound our estimated effect of a connection.[17] We estimate unweighted regressions and check that our results are robust to heterogeneous effects by age or immigrant cohort. Standard errors are clustered at the level of the displaced individual i, as the outcome can be equal to one for at most one potential hiring firm k, implying that hiring outcomes are not independent across different values of ik for a given individual. Our main specification includes 1,646,638 potential hiring firms k, 68,601,502 ik dyads, and 3,330,436 firm pairs jk. The median jk firm pair contains 8 individuals.

4.1.2 Discussion

The credibility of our design depends on whether workers displaced by the same closure who differ in connectedness to firm k are otherwise comparable, conditional on controls and fixed

[16] This restriction also reduces the computational burden: otherwise the dyadic dataset, with cardinality $|I| \times |K|$, would be intractably large.

[17] The restriction applies to 0.4 per cent of native worker-firm dyads and 0.7 per cent of immigrant worker-firm dyads where a connection of some type exists. We also show in Figure A.4 that both recent and established immigrants' connections are only weakly affected by this restriction, even if the tendency to have connections of more than one type does grow with time in Sweden.

effects. Under this assumption, Equation (1) recovers the causal effect of a connection on hiring outcomes. There are two main threats to this assumption. First, some individuals might be more likely both to have connections and to get hired after a displacement. They may be more employable or better at searching and may have previously worked at larger firms or had more job transitions. Second, connectedness may correlate with "typical" mobility patterns for certain types of workers between certain types of firms, due e.g. to industry or occupational persistence or workplace segregation. Failure to adequately control for such determinants of employability or mobility would bias our results.

Our fixed effects already ensure that we compare similar workers, and absorb any factors that vary across closing firms, such as last industry or differences in the local business cycle. Our controls X_{ik} include a rich set of worker–firm covariates, which aim to measure the type of worker that firm k typically hires: i's region of origin, prior work experience in firm k's industry and municipality, workforce composition at firm k, and the nativity of firm k's manager (Åslund et al., 2014). Systematic differences in hiring across firm pairs, due e.g. to industry differences, are absorbed by firm-pair fixed effects α_{jk}.

Nevertheless, it is in principle still possible that having a connection of a given type at hiring firm k correlates with unobserved determinants of the probability of a market hire. We therefore pursue two approaches to mitigate such concerns. First, we allow the displacement-event fixed effects α_{jk} to vary across groups defined by prior work experience, nativity, education, pre-displacement earnings (including interactions), or occupation. In each case, identification will come from comparisons between workers with broadly similar employment histories or observed skill profiles within the same closing firm. Second, in Section 4.2.2, we assess the possibility that estimated connection effects might capture typical mobility patterns for certain types of workers between certain types of firms, rather than genuine network links, through a placebo analysis based on "connections that could have been but were not." We find placebo effects an order of magnitude smaller than our true effects, supporting the interpretation of our main estimates as capturing network effects.

The constant-effects model outlined above also makes an implicit "no spillovers" assumption analogous to the standard Stable Unit Treatment Value Assumption (SUTVA). Specifically, the model assumes that the effect of a connection depends neither on which other firms a displaced individual is connected to, nor on which firms their co-displaced coworkers are connected to. We discuss this assumption further in Appendix B and present supporting evidence. We also check the robustness of our results to less demanding assumptions by allowing the effect of different types of connections to depend on the size of an individual's network.

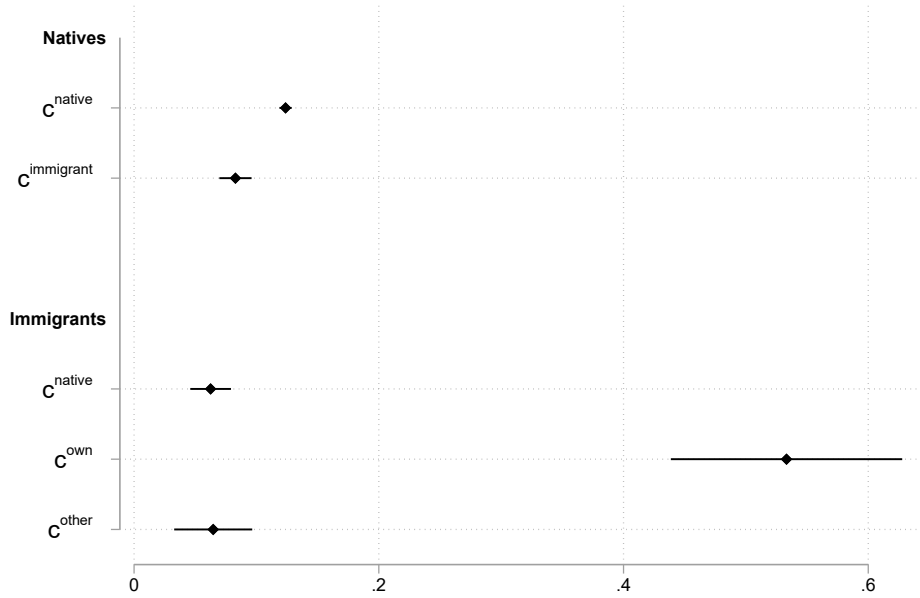
Finally, focusing on displaced workers makes the decision to search for work exogenous to the size and composition of a worker's network. However, selective return migration could still bias estimates if displaced immigrants with weaker networks are more likely to leave Sweden, rather than search for another job. Table 1 shows, however, that post-displacement emigration rates are modest—about 4 per cent within the first five years after arrival and below 1 per cent thereafter—suggesting limited scope for this bias.

4.2 Hiring at connected firms

4.2.1 Main results

Figure 3 presents our main estimates of the impact of having a connection at a given firm on the probability of being hired there. There are two main messages: (i) connections matter; (ii) the impact of a connection depends on the origin of the displaced worker and the connection. The estimates suggest that for displaced natives, having a native connection increases the probability of being hired at a given destination firm by 0.12 percentage points, an economically significant effect. For immigrants, on the other hand, conational connections increase the hiring probability by over 0.5 percentage points while native connections only increase the hiring probability by 0.06 percentage points. By way of comparison, the baseline probability that an unconnected worker is hired by any of the K firms included in the sample is, on average 0.023 per cent. Consistent with homophily in the value of connections, immigrant connections matter less than native connections for natives, and other-group connections have a smaller impact for immigrants.

Figure 3: Effect of a connection for displaced workers



Notes: Effect of a connection on the probability a displaced immigrant is hired by firm k, estimated using Equation (1). The dependent variable, an indicator for being hired by firm k, is scaled by 100. Standard errors are clustered by displaced individual i, 95 per cent confidence intervals are reported.

Table 2 investigates the robustness of our identifying assumption; column 2 contains the main estimates from Figure 3. In column 1 we omit the controls, reporting simple within-dyad (closing firm j by hiring firm k) differences between connected and unconnected individuals. The controls capture potential confounds that could increase both the likelihood of having a connection to firm k and the probability of being hired by firm k: origin similarity with the manager and/or the workforce, and prior work experience in the industry or location of firm k.

The point estimates decrease slightly (from column 1 to 2), but are not materially affected by the inclusion of the covariates.

The identifying assumption that connections are conditionally randomly assigned within closing firm j may be violated if individuals with a stronger attachment to the labour market or more connections are in general more likely to be hired, even conditional on our controls. In column 3, we further restrict ourselves to comparisons of workers who have the same number of prior workplaces (workers with zero or one past workplace are grouped together). In columns 4 and 5 we instead define deciles of, respectively, the distribution of past coworkers (from prior workplaces) or the distribution of connected workplaces, and only compare workers with a similar number of past coworkers (column 4) or connected workplaces (column 5). All coefficients are remarkably stable, with the exception of the effect of other immigrant connections for displaced immigrants, which becomes insignificant when comparing individuals with the same number of connected workplaces.

Table 2: Effects of connections on firm hiring

	(1) Hired by k	(2) Hired by k	(3) Hired by k	(4) Hired by k	(5) Hired by k	(6) Hired by k	(7) Hired by k	(8) Hired by k
Natives								
c^{native}	0.14	0.12	0.12	0.12	0.12	0.091	0.12	0.13
	(0.002)	(0.003)	(0.003)	(0.004)	(0.004)	(0.003)	(0.006)	(0.006)
$c^{immigrant}$	0.095	0.083	0.079	0.090	0.090	0.070	0.12	0.087
	(0.007)	(0.007)	(0.008)	(0.010)	(0.010)	(0.009)	(0.02)	(0.01)
Immigrants								
c^{native}	0.077	0.063	0.066	0.083	0.055	0.031	0.051	0.049
	(0.008)	(0.008)	(0.01)	(0.01)	(0.01)	(0.02)	(0.04)	(0.02)
c^{own}	0.58	0.53	0.50	0.62	0.62	0.45	0.69	0.50
	(0.05)	(0.05)	(0.06)	(0.07)	(0.08)	(0.08)	(0.2)	(0.1)
c^{other}	0.082	0.065	0.063	0.080	0.0091	0.081	0.069	0.022
	(0.02)	(0.02)	(0.02)	(0.03)	(0.03)	(0.04)	(0.07)	(0.03)
Controls	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fixed Effects	–	–	PW	PC	CW	PC, IM, N	PC, Ed × ET, N	Occ
$\hat{\mathrm{E}}[Y(0)\|c^{any}=0]$	0.023	0.023	0.022	0.021	0.021	0.019	0.017	0.029
N (000s)	69,554	68,601	65,068	60,203	60,415	57,485	40,984	16,810

Note: The dependent variable, an indicator for being hired by firm k, is scaled by 100. Controls are share native, conational, or same education as i, indicators for i having worked in same industry or municipality as k, all interacted with an indicator for i's nativity, categories for region of birth and education, and the main effects and interaction of i's nativity and the nativity of k's manager. All specifications include a jk = firm pair fixed effect. The jk fixed effect may be interacted with the following categorical variables: PW = number of past workplaces (five categories); PC = number of past coworkers (ten categories); CW = number of connected workplaces (ten categories) , IM = previously worked in k's industry and municipality; N = nativity (two categories); ET = earnings tercile in j; Ed = education (three categories); Occ = 2-digit occupation (SSYK96). Standard errors clustered by displaced individual are reported in parentheses.

In columns 6–8, we address the possibility that estimated connection effects partly reflect systematic sorting across industries, locations, or occupations rather than network effects per se. In column 6, we add an interaction between the fixed effect α_{jk} and an indicator for whether worker i's last prior job was in the same industry and municipality as firm k. In column 7, we instead restrict ourselves to comparisons of workers with the same education level interacted with earnings tercile in the closing firm. By comparing workers with similar observed skill profiles within the same closing firm, this specification addresses the possibility that nationality-based connections simply proxy for similarity in specialisation. Both columns 6 and 7 include a jk by nativity fixed effect, eliminating potential confounding due to immigrant-native segregation across firms. Finally, in column 8, we interact the jk fixed effect with the displaced worker's 2-digit occupation in the closing firm.[18] These refinements sometimes reduce the point estimates somewhat, particularly when considering occupation, but they do not alter the main qualitative pattern: own-group connections remain much more important for displaced immigrants than native connections.

The causal effect of a connection may also vary systematically by other displaced worker characteristics beyond nativity. Figure A.5 reports results where we interact the connectedness indicators with gender, education, age, region of birth, past turnover, or network size. The finding that own-group connections matter much more holds for all subgroups. For natives, the effect of a connection tends to be greater for males than for females, decreases with the level of education and increases with age. These differences are similar though less clear-cut for immigrants. Own-group connections are more influential for immigrants born in Eastern Europe or in Non-European countries than for those born in Western Europe and North America.[19] Higher past job turnover (i.e. more past job transitions) appears not to affect the estimated effects. However, we do find that all types of connections have systematically larger effects for individuals connected to fewer firms, suggesting that relaxing the no-spillover assumption may be relevant. We return to this point in Section 4.2.3.

Finally, we confirm in Table A.4 that our results are robust to alternative sets of connections and sample definitions. In column 2 we only include connections through coworkers in the most recent workplace, increasing the estimated effects. In column 3 we include individuals who have recently worked in a firm without well-defined workplaces in our sample, while in column 4 we randomly sample 100 coworkers in past workplaces with more than 100 employees and include connected firms based on these sampled past coworkers in our main sample. Neither change materially affects our estimated effects. Dropping individuals with no connections at all (column 5), or dropping segregated closing workplaces (column 6), does not have a significant

[18] There are three main reasons our sample is substantially smaller in this case: (i) occupation is only observed from 2001 onward; (ii) occupation is recorded in survey, rather than administrative data, and interpolated where possible, leading to missing values; and (iii) the analysis requires variation in connectedness within closing/hiring-firm by occupation cells.

[19] Our grouping into Western Europe, Eastern Europe, and Rest of World serves as a rough proxy for migration motives: Western Europe is mainly labour migrants, the Rest of World category consists mainly of humanitarian and their families, whereas Eastern Europe combines labour migrants from recent decades and historic refugee migration.

effect on the estimated coefficients. Clustering standard errors by hiring firm k (column 7) similarly does not alter our conclusions.

4.2.2 Placebo analysis

Systematic transitions between certain types of firms or unobserved similarities between displaced workers and their past coworkers, making them both attractive to the hiring firm, could bias the estimated connection effects. To probe these threats, we conduct a placebo analysis in which we repeat the main analysis using simulated connections drawn from comparable but randomly selected contexts. The intuition is straightforward: if our estimates simply captures systematic mobility patterns between firm types due to sorting on either unobserved types or accumulated local industry-specific human capital, then similar "effects" should also appear for placebo connections. In addition to shedding light on causality per se, the exercise can also illuminate whether the unobserved factors affect the relative size of conational versus other types of connections.

For each displaced worker (with a non-zero network in the displacement year t, who has not recently worked at a firm without defined workplaces), we identify the municipality and two-digit industry of all firms where they interacted with a past coworker. For each past coworker, we randomly draw (without replacement) another individual of the same nativity, employed at a firm with 100 or fewer workers in the same local industry and year as the displaced worker's true past firm. The set of randomly drawn individuals are the displaced individual's "placebo past coworkers".[20]

We estimate the same set of dyadic specifications using our placebo sample as we did for the true sample; see Table A.5 for estimates. Column 1 reports our estimates of the effect of a true connection, while columns 2–4 report the effects of placebo connections. A first message is that the estimates for actual connections are much larger than for the corresponding placebo connections. Thus, the main component of our baseline estimates does seem to reflect a positive causal impact of connections. Nevertheless, the placebo estimates, while small, are statistically significant. We believe this is expected for two reasons. First, there is likely to be some degree of unobserved similarity and correlated movement among workers in the same local industry, which these estimates pick up. Second, some of the "placebo impact" arguably captures broader industry-level network effects from workers in the same local industry interacting across firms and then making use of such connections following displacement. This would lead us to conclude that we may underestimate the true influence of professional networks.

[20] We use the same procedure to construct placebo connections from the placebo coworkers as we used with the true coworkers. We identify their workplace in year t, and say that the displaced worker has a placebo connection if the placebo coworker is working in a firm that is neither the closing workplace, nor the displaced worker's past workplace (where they interacted with their true past coworkers). We restrict the set of potential placebo past coworkers to exclude true coworkers from different years.

4.2.3 Nativity and tie characteristics

The results presented thus far indicate that homophily based on country of birth is a powerful predictor of the impact of a connection on the likelihood of a displaced worker being hired by a particular firm. We now explore whether this reflects homophily per se rather than other tie characteristics.

First, we assess whether homophily by nativity is merely a proxy for similarity in other dimensions by further categorizing connections based on shared gender, education level, and age. Figure A.6 displays the results alongside the original estimates. The probability of a match generally increases when the displaced worker and the connecting worker share a demographic trait. But homophily in other dimensions is of secondary importance compared to homophily in country of birth. For example, native connections with the same education level increase the hiring probability for displaced immigrants by 0.06 percentage points, whereas conational connections with different education levels boost this probability by 0.49 percentage points.

Next, Figure A.7 examines whether the results are due to native versus conational connections differing in terms of some measure of tie quality. Conational connections might be less likely to leave their firm between t, when connections are measured, and $t+1$, by which time the closure has occurred. In Figure A.7a, we differentiate between coworkers who remain employed at the hiring firm by $t+1$ and those who do not. The results show that connections who are still employed have a much larger impact on hiring, regardless of nativity.[21]

Displaced immigrants tend to come from smaller workplaces (median of 11 coworkers) compared to natives (median of 13). Conational connections may be more valuable because they were formed in such establishments, where interaction tends to be more intensive. Figure A.7b demonstrates that connections formed in workplaces with 20 or fewer coworkers are indeed more valuable, but this effect holds across all types of displaced workers and connections.[22]

We also consider how relative positions in the earnings distribution affect the value of a connection (Figure A.7c). Point estimates are consistently somewhat greater for connections through a former colleague of higher earnings rank. But it is only in the case of native connections for displaced natives that the difference in the estimates is statistically significant. Figure A.7d illustrates that connections from the two years preceding displacement (relative to older connections) are more valuable. In a similar vein, longer interactions in the past workplace tend to yield stronger effects (Figure A.7e) and connections to expanding firms are more beneficial (Figure A.7f). However, in all cases, the differences we document are consistent across both displaced worker and connection types.

Overall, connections that we would intuitively expect to be more influential are indeed associated with stronger hiring effects. Nevertheless, conational connections remain substantially

[21] This result also lends support to our interpretation of the effect of connections as reflecting information transmission, rather than unobserved similarity between the displaced worker and their past coworker. If the latter were the case, we should observe a similar effect of a connection on the probability of a hire, regardless of whether the past coworker continues to be employed at the hiring firm.

[22] The cutoff of 20 is based on previous evidence suggesting that the value of connections decays more rapidly above this threshold (Eliason et al., 2023)

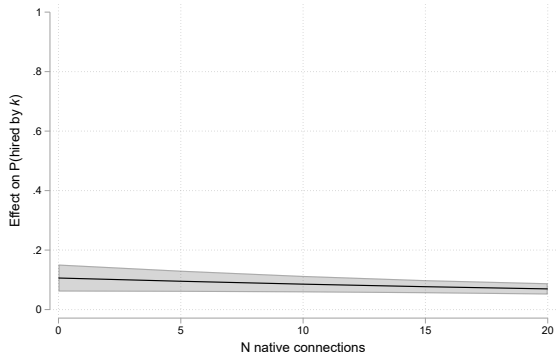
more valuable than native connections also conditional on other dimensions of tie quality. Furthermore, the excess influence of origin similarity on the probability of being hired is in most cases much greater than that of other measures of tie quality.

Finally, as shown in Figure A.5f, the effect of a connection depends somewhat on the size of a worker's network. It may also be that the effect of a connection declines with additional connections *of that type*. Since immigrants have fewer conational than native connections, differences in network size may partly explain their differential effects. To examine this, Figure 4 presents estimates from a specification where we interact our connectedness indicators with the total number of connections of that type. The downward-sloping lines suggest decreasing marginal effects in particular for conational and other immigrant connections. The differences are, though, only statistically significant when comparing extremes of the range of the number of connections.

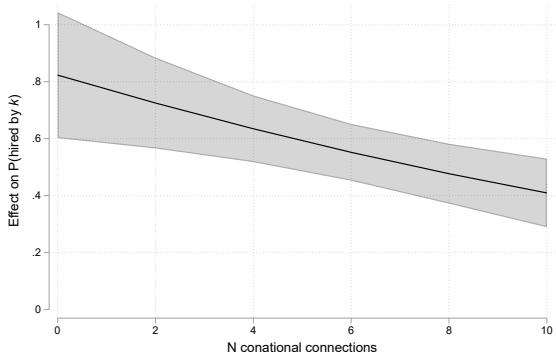
Taken together, the evidence in Figure 4 suggests that the assumption of a constant marginal effect of a connection, implicit in Equation (1), appears reasonable for the number of connections typically observed in our sample (in particular, having more than five own-group connections is uncommon). Although the marginal effect of a connection declines somewhat with the number of same-type ties, the marginal effect of a conational connection remains much larger than that of a native connection even when holding constant the number of ties of each type. However, because the marginal value of an additional native connection is almost constant, there is more scope for the contribution of native ties to re-employment to continue to grow even as immigrants accumulate many more such connections. We will return to relative importance of accumulating few high-impact conational connections versus many low-impact native connections in Section 5.

Figure 4: Effect for displaced immigrant by number of connections

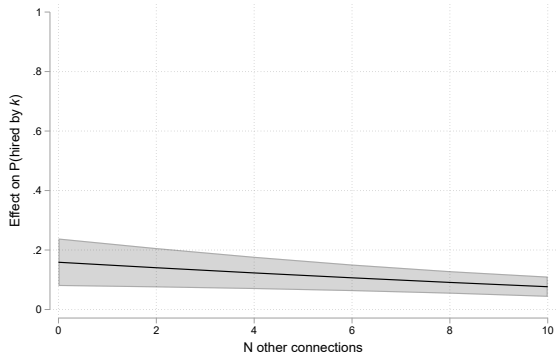
(a) Native connections



(b) Conational connections



(c) Other immigrant connections

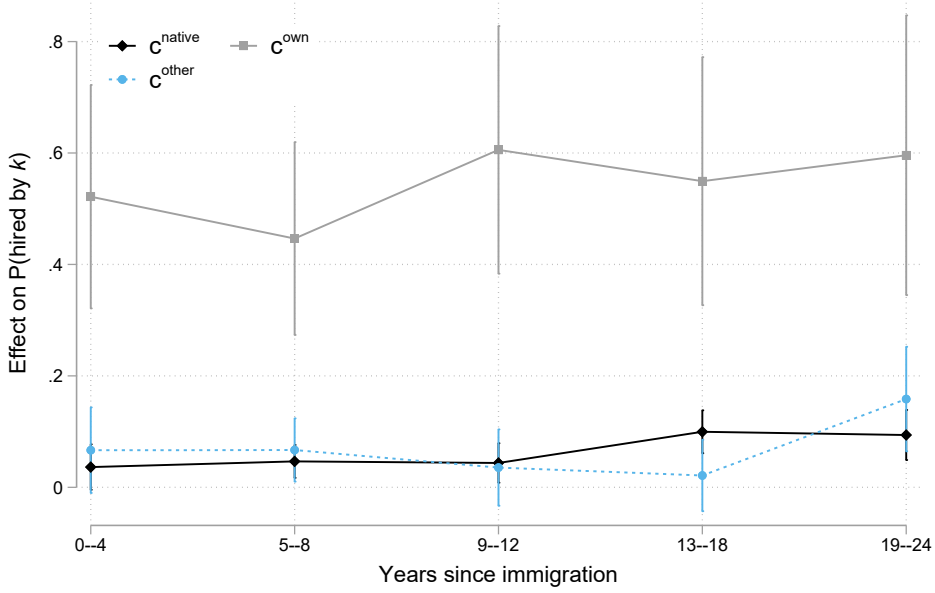


Notes: Heterogeneous effects of connections on the probability a displaced immigrant is hired by firm k, estimated by interacting the connectedness indicators in Equation (1) with the total number of connections of that type, including main effects. Standard errors are clustered by individual i, 95 per cent confidence intervals are reported.

4.3 Time-varying effects of firm connections

Figure 2 documented that the re-employment probability for immigrants increases with time in Sweden and suggested that this might be linked to increases in the size of immigrant networks. Another possibility is that immigrants' networks, in particular native connections, become more productive over time in Sweden as immigrants acquire complementary host country-specific human capital. To explore this possibility, we interact the connectedness indicators for displaced immigrants in Equation (1) with a set of indicators for years since immigration.[23] This analysis comes with certain caveats. First, we do not estimate the causal effect of years since immigration on the effect of a connection, but rather how years since immigration predicts variation in the causal effect of connections of different types. Second, our sample is a pooled cross-section; we do not observe individuals displaced at multiple points in time since immigration (which would be a small and strongly selected sample). As is common in analysis of immigrant assimilation, selective return migration may therefore explain part of the observed dynamic patterns.

Figure 5: Effects for displaced immigrants by years since immigration



Notes: Heterogeneous effects of connections on the probability a displaced immigrant is hired by firm k (scaled by 100), estimated by interacting the connectedness indicators for displaced immigrants with a cubic polynomial in years since migration at the time of displacement, including main effects. Marginal effects of a connection for selected values of years in Sweden are shown. Standard errors are clustered by displaced individual i, 95 per cent confidence intervals are reported.

The effect of a connection on the hiring probability is plotted in Figure 5. The estimates show that the higher productivity of own-group connections persists for at least two decades. The effect of native connections does, however, differ substantially between those displaced at shorter and longer durations. Rising from around 0.04 percentage points in the first four years

[23] We also include the main effects of the years since immigration dummies.

to 0.10 percentage points among those residing 13–24 years in Sweden means that most of the gap to the effect of native connections for natives is closed (cf. the point estimate of 0.12 of column 2 of Table 2).

Earlier arrival cohorts will be more heavily represented among displaced immigrants with a long tenure in Sweden. We check that the time profile of effects is not capturing cohort effects by interacting the firm-dyad fixed effect with a cohort fixed effect.[24] Estimates from these within-cohort analyses are shown in Figure A.8. The growth in the effect of native connections is roughly similar to the original specification, where we compared individuals both within and across cohorts, going from 0 in the first years after arrival to 0.08 percentage points after 13–24 years.

To what extent does the change in the effect of a native connection reflect true changes in the social interactions of immigrants and natives over time? To answer this question, we study how our measures of tie quality evolve with years in Sweden for different types of connection, in Figure A.9. Some indicators of tie quality, such as the size of the workplace where the interaction took place, or the relative earnings, are relatively stable over time since migration. The main dimension along which ties appear to improve in quality is that they have spent longer interacting in the prior workplace. However, this is true for all connection types, not native connections specifically. There is also some evidence of ties becoming of worse quality in other dimensions: interactions become less recent with years in Sweden, and past coworkers become more dissimilar in age. Taken together, these results provide no evidence that native connections become higher-quality than other connections over time. Rather, the increased effect likely reflects changes in how immigrants and natives interact—either in the former workplace or during job search

4.4 Hires in local industries, earnings and turnover

So far we have investigated and established the causal effect of connections on the probability that a displaced worker is hired by a given firm. However, hiring by connected firms may not fully capture all the ways displaced workers use their networks to find a job. To present a more complete picture of how immigrants use their professional connections to find work, we now turn to analysing how connections are associated with other outcomes of displaced workers' job search.

4.4.1 Hires in a connected local industry

Past coworkers may provide information about vacancies at other firms where they are not currently employed. We therefore extend the dyadic approach to consider how industry-level connections via past coworkers increase the probability of a displaced worker being hired by a firm in that industry. For each closing firm j, we identify the set of local industries that are

[24] We define the following cohorts by first immigration date: prior to 1980, 1980–1989, 1990–1999, from 2000; natives are considered a separate cohort.

connected to the closing firm through a former coworker of a displaced individual.[25]

For each displaced worker, we construct a dyadic observation for each potential hiring local industry and define indicator variables for (i) whether the displaced worker is hired by the local industry; and (ii) whether they have a connection to the industry via a former coworker. We present some summary statistics on the individual-local industry dyads in Table A.6. Displaced individuals are only connected to a very small fraction of firms in a given local industry; in 5–10 per cent of cases, individuals are connected to more than one firm in a potential hiring industry.

We estimate a linear probability model on the local industry dyadic data, analogous to Equation (1), to recover the effect of a local industry connection on the hiring probability. The vector of controls includes dummies for the region of origin of the displaced worker, as well as indicators for whether the worker previously worked in the two-digit industry, municipality, or the interaction of industry and municipality defining the local industry k. The model also includes closing firm j by local industry fixed effects. As in the firm-level analysis, the effect of a connection is therefore identified from comparisons among workers displaced from the same closing firm and facing the same potential destination local industry, but who differ in whether they have a connection to that industry.[26] Since industry-level connections are more likely to reflect occupational or job-type persistence than firm-level connections, we caution against a causal interpretation of these estimates, in contrast to our firm-dyadic analysis. In particular, workers connected to a local industry through past coworkers may simply be more likely to search in that industry or possess industry-specific skills that make them more likely to be hired there.

In Figure 6 we first report the effect of having a connection in a local industry on the probability of being hired in that industry ("All industry hires"). We then decompose the total effect of a connection on industry hiring into the effect on hiring at the subset of firms in the local industry where a former coworker is working ("Direct connection") and the residual set of firms in the local industry where no former coworker is working ("Indirect connection"). The indirect component captures whether past coworkers transmit information about vacancies elsewhere in the local industry. For comparison, we also report the effect of a connection on firm-level hiring, using the same individuals as we used to estimate the effect of connections on industry hiring ("Firm hires").[27]

For displaced natives', connections have a significantly larger effect on the probability of being hired by a local industry compared to what is revealed in the firm-level analysis. This is particularly true of their native connections, where the respective estimates are 0.11 for firm connections and 0.31 percentage points for industry connections. The positive "Indirect con-

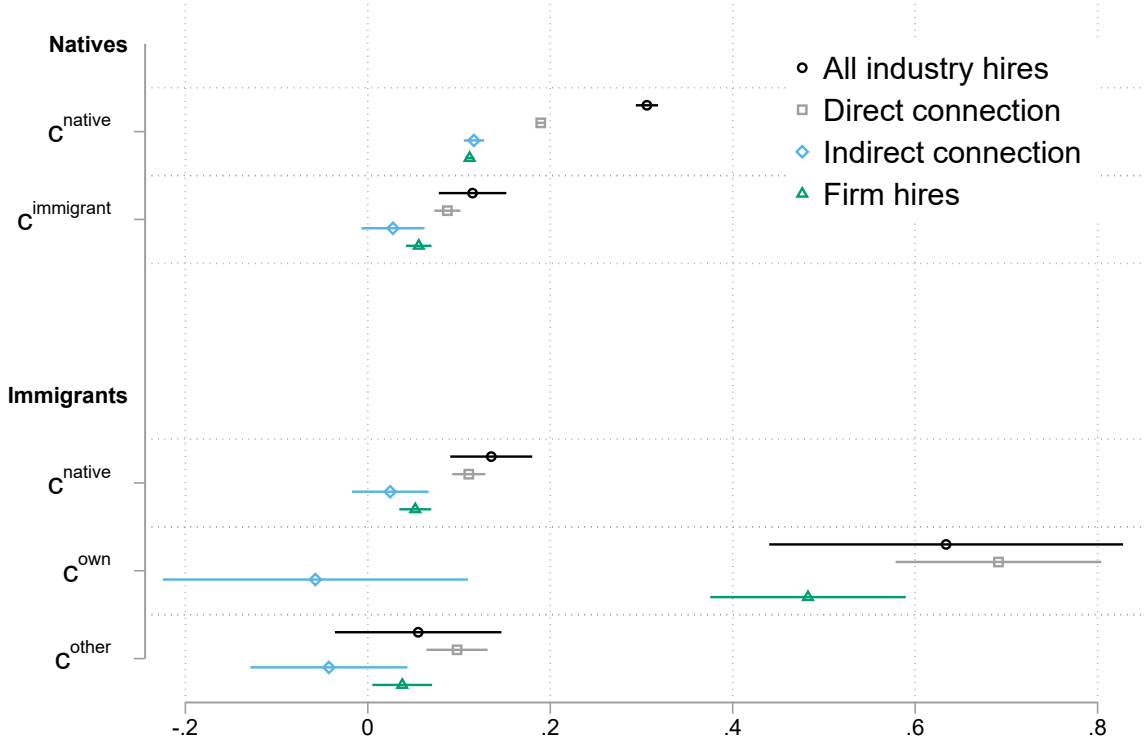
[25] Local industries are defined as the set of firms in the same two-digit industry and municipality.

[26] We here drop individuals who are connected to local industry k by more than one type of connection, leading to a smaller sample than when considering firm hiring, and cluster standard errors by closing firm j by local industry k.

[27] The comparison group in our industry estimation includes only those individuals without any connection to the local industry, whereas in the main firm analysis it additionally includes individuals with connections to the local industry but at another firm than the hiring firm k.

nection" estimate furthermore suggests that natives' (native) connections do appear to provide information beyond vacancies in their own firm. For displaced immigrants, however, indirect connections do not seem to matter regardless of type. Thus, connections improving opportunities beyond firm borders does not seem to be a factor benefiting immigrant workers.[28]

Figure 6: Effect of a connection on hires by firms and local industries



Notes: Coefficients estimated using Equation (1). Standard errors are clustered by displaced individual i; 95 per cent confidence intervals are reported. Details for the regression are available in Table A.7.

4.4.2 Earnings and turnover associations

To understand whether homophily affects the type of information a connection typically shares, and whether connected hires in local industries reflect different types of information flows than connected hires in firms, we now study how the wages and turnover of connected hires differ from those of unconnected hires. Dustmann et al. (2016a) and Glitz and Vejlin (2021) analyse a model of referral hires, where referrals lower the uncertainty to the employer about the productivity of a match with the referred worker. As a result, starting wages are higher for referred workers than

[28] While we limit attention to hiring local industries where a worker has at most one connection type, workers may have multiple connections (of the same type) to different firms in the same local industry. In column 4 of Table A.7 we restrict the sample to industries where the displaced connection has at most one past coworker working. Coefficients are smaller in this case, however, we do still find evidence of larger effects for natives'.

for non-referred workers. The model also predicts lower turnover, since the firm makes fewer wrong decisions in hiring workers. However, over time, initial uncertainty about the unobserved productivity of workers is resolved. Bad matches are endogenously terminated and productive workers hired without a referral are able to renegotiate their wages up. As a result, initial differences between referred and non-referred workers fade over time.

To test whether certain connections provide referrals (versus vacancy information), we consider the sample of workers i who were displaced by the closure of j and who are employed at firm k in November of year $t+1$. We construct a panel of outcomes for these workers over the years $\tau \in [t+1; t+5]$ and estimate versions of the following equation, following Glitz and Vejlin (2021):

$$y_{ijk\tau} = \alpha_j + \sum_{s=1}^{5} \gamma_s c_{ik} \mathbf{1}(\tau = t+s) + \sum_{s=1}^{5} \beta_s \mathbf{1}(\tau = t+s) + X'_{ijk\tau}\beta + \epsilon_{ijk\tau}. \tag{2}$$

The outcomes we consider are log annual earnings in year τ and an indicator for leaving one's current employer between November of year τ and November of $\tau+1$.[29] Importantly, we do not condition on continued employment at firm k after $t+1$. c_{ik} is an indicator for hired worker i having a connection to hiring firm k of some sort; we will study the effect of connections through past coworkers to both the hiring firm and other firms in the same local industry and we will allow the effect of connections to vary by the degree of homophily between the hired worker and their past coworker. We include a closing firm j fixed effect and an extensive set of pre-displacement demographic and employment history covariates $X_{ijk\tau}$. In particular, the set of covariates includes a quadratic in labour earnings in year t, i.e. pre-displacement, as well as indicators for having recently worked in either firm k or the two-digit industry of firm k, and for having the same nativity as the manager at firm k. The effect of these covariates is allowed to vary with time since initial hire, τ. A causal interpretation of this analysis is subject to caution, since the sample selection necessarily conditions on displaced workers having found a job by November of $t+1$. However, such an analysis still provides valuable information on whether connections referred the displaced worker to the hiring firm or not.

Figure 7 presents results disaggregated by homophily in nativity between the hired worker and their connection at the hiring firm. Whether our estimated effects align with the theoretical predictions turns out to depend crucially on whether there is homophily in nativity between the displaced worker and their connection. Displaced natives hired through native connections have higher earnings, particularly at entry, and lower turnover. Although somewhat noisy, we see a similar pattern for displaced immigrants, who have higher earnings and lower turnover when hired through an immigrant connection (irrespective of whether an own- or other-nationality connection). The time pattern of effects—a short-term increase in earnings and a decrease in turnover—is consistent with homophilous connections providing referrals. Neither group appears to benefit strongly from non-homophilous connections upon hire.

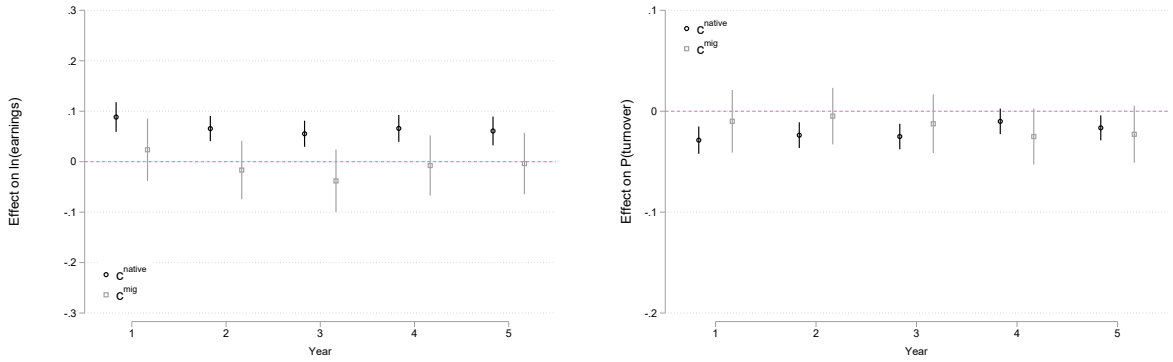
Further analysis suggests that if one aggregates all types of connections, there are no signif-

[29] Hourly wages or monthly hours worked are only observed for a subset of workers, so we study earnings rather than wages.

Figure 7: Labour market effects of firm connections

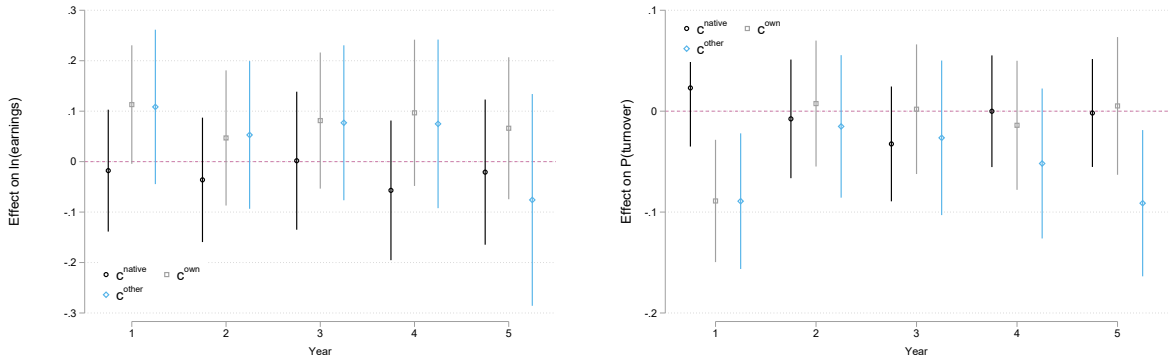
(a) Displaced natives, earnings

(b) Displaced natives, turnover



(c) Displaced immigrants, earnings

(d) Displaced immigrants, turnover



Notes: Coefficients estimated using Equation (2), allowing the effect of a connection at the hiring firm on subsequent outcomes to vary by the nativity of the displaced and connecting workers. Standard errors are clustered by closing firm j and 95 per cent confidence intervals are reported.

icant differences between the associations found for displaced immigrants compared to natives (Table A.8). We also find that having connections elsewhere in the industry where hired has only a small positive association with earnings and none with turnover, suggesting that industry connections are providing information about vacancies, rather than referrals (Figure A.10).

Finally, we also investigated how the effect of a native connection varies with years since immigration, measured at the time of the displacement (Figure A.11). We estimate the largest associations between having a native connection and earnings or turnover for immigrants who have been in Sweden the longest, consistent with immigrants becoming more likely to receive native referrals with longer host country durations. However, these estimates are imprecise, particularly for earnings, and we cannot reject that there is no change in the association between having a native connection and earnings or turnover with years in Sweden.[30]

5 Networks and assimilation profiles

5.1 Networks and re-employment

We have shown that immigrants' networks expand and shift in composition over time and that displaced workers benefit from these connections, with effects varying by connection type and time spent in the host country. To evaluate the economic significance of these findings, we quantify the total contribution of network size, composition, and effectiveness to the re-employment of immigrants and natives.

We calculate the contribution of networks to the re-hiring probability for a given type of displaced worker in the following way: (i) multiply the effect of a connection of a given type (reported in Section 4) by the average number of connections held by that type of worker (reported in Figure 2a); (ii) sum over connection types (see Appendix C for details). We then use the corresponding estimates for natives to create age-weighted and years-since-migration specific comparisons.[31] The exercise helps to uncover three main channels for changes in the total productivity of networks: (i) changing number of connections; (ii) changes in the composition of connections; (iii) changes in the productivity of each type of connection.

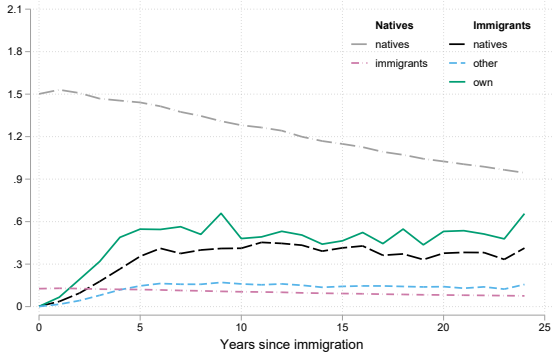
Figure 8 reports the contribution of native and immigrant connections to the probability of re-employment for immigrants and similar-aged natives. When assuming a constant effect of a connection (8a), the estimated contribution of the respective types of connections increase in the years when immigrants enter the labour market and start accumulating workplace-based contacts. Allowing the effects of a connection to vary with years since migration (8b) illustrates how the growing importance of native ties over time reflects not only their rapid accumulation, but

[30] It is of course also possible that there are countervailing mechanisms. For example, workers with longer durations not relying on connections may be more likely to re-optimise based on acquired country-specific skills, leading to better matches when re-employed.

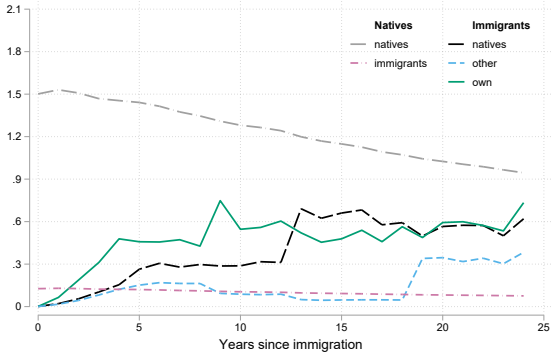
[31] The procedure is to multiply the age-invariant effect of natives' connections by the age-weighted average number of connections of that type for each value of years since migration, using the same re-weighting procedure as in Figure 2a.

Figure 8: Contribution of networks to assimilation in re-employment after displacement

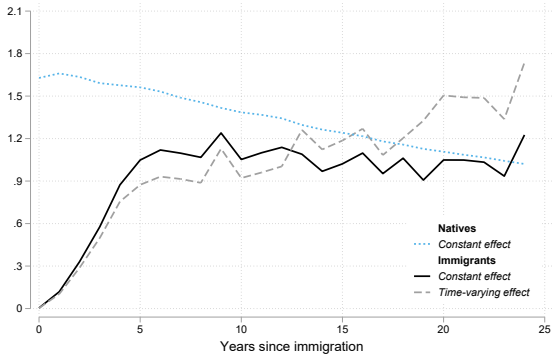
(a) Constant effects, by type



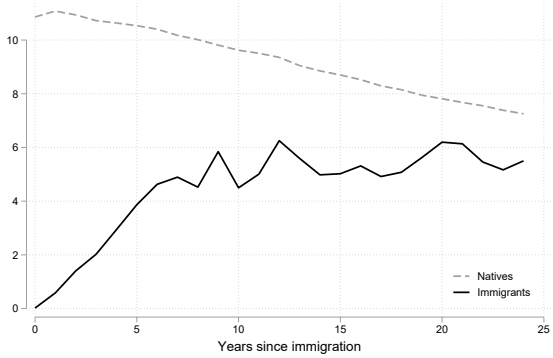
(b) Time-varying effects, by type



(c) Total effect, only small workplaces



(d) Total effect, all workplaces



Notes: Contribution of networks to re-employment in November of $t+1$, defined according to Equation (C.2). Both constant (time invariant) and varying-by-years-in-Sweden estimates of the effect of a connection are considered. The contribution of natives' networks to their re-employment probability is calculated using the number of connections of different types for similar-aged natives for a given number of years since immigration. Note the change in y-axis scale when including all past coworkers.

also an increase in their productivity. The figure helps distinguish between two sources of labour market integration: early improvements driven mainly by the accumulation of rarer but high-productivity conational ties, and later improvements driven by changes in the effectiveness of more-common native connections.[32] In terms of immigrant-native convergence, we see a narrowing of the (age-adjusted) gap in the contribution of native connections, and a slightly increasing excess role of foreign-born (cf. own + other vs. immigrants) connections for immigrants.

Using our baseline network definition, the estimated total effect of networks on the re-employment probability of immigrants and natives is modest in absolute terms (Figure 8c). For example, focusing on immigrants displaced after 10 years in Sweden, the contribution of connections to re-employment is 0.9–1.1 percentage points. For similar-aged natives the figure

[32]Precision decreases with the time-varying estimates, urging some caution in interpretation. The late and sharp increase in the contribution of other-migrant connections may partly be a product of this uncertainty.

is about 1.4 percentage points. Relative to the 6 percentage point increase in re-employment among immigrants observed over the first ten years, the network contribution can be seen as somewhat more significant. In the long run, we also see that the contribution of immigrants' connections converges to, or even surpasses, the contribution of natives' connections.[33]

Figure 8d presents results based on a broader network measure that also includes connections formed in larger workplaces. Here we multiply the average number of connections for immigrants displaced at different durations in Sweden by connection effects estimated on a sample that also includes connections to firms through from a random sample of past coworkers from larger workplaces (see Table A.4, column 4). We then see much larger total contributions. Similar-aged natives' networks increase their re-employment probability by nearly 11 percentage points compared to new immigrants, while the contribution of immigrants' networks to their re-employment rises to 5–6 percentage points after 10–20 years in Sweden. Using this broader measure, native connections become more important. As native ties are more common when coworkers from larger past workplaces are included (Figure A.13), their contribution exceeds that of conational connections after only three years in Sweden, and continues to grow until immigrants have been in Sweden for around 12 years.

5.2 Networks and unemployment in equilibrium

Immigrant assimilation is often assessed by how quickly immigrant employment or earnings converge toward native levels. To relate our micro-level estimates of network effects on re-employment to such aggregate outcomes, we use a simple steady-state framework which is standard in search models of the labour market. The aim is to translate observed job-finding rates into implied unemployment rates for each group.

In steady state, the unemployment rate for group g depends on the rate at which workers lose jobs (δ_g) and the rate at which they find new ones (λ_g):

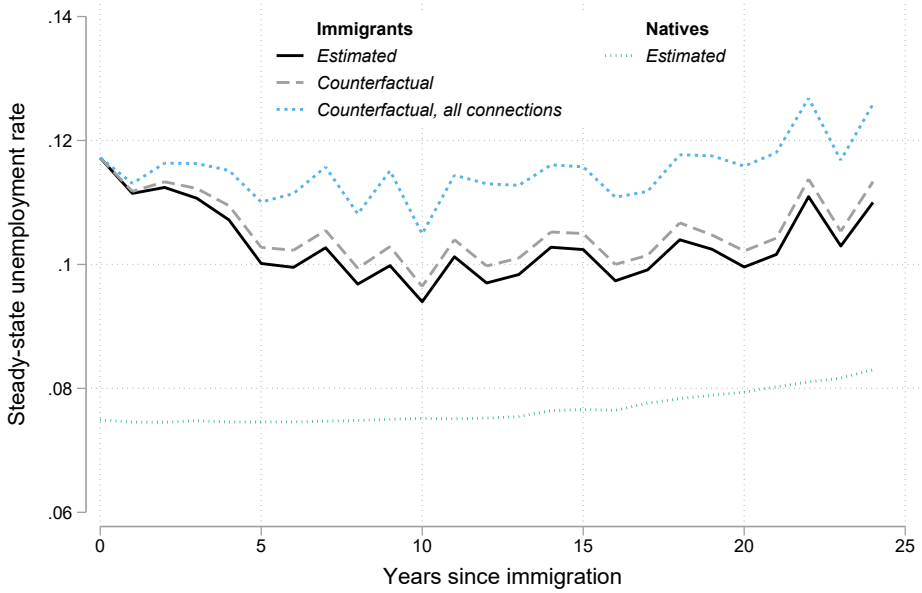
$$U_g = \frac{\delta_g}{\delta_g + \lambda_g}. \tag{3}$$

We interpret the observed one-year re-employment probabilities in Figure 2b as annual job-finding probabilities, assuming that job arrivals follow a Poisson process. For example, displaced immigrants who have just arrived in Sweden face an implied monthly job-arrival rate of $-\frac{1}{12}\ln(1-0.557) = 6.8$ per cent, compared with 11.1 per cent for similar-aged natives. To abstract from potential differences in job separations, we assume a common monthly separation

[33]The total contribution of same-aged natives' networks declines as older natives, who are less mobile and work in smaller firms, have fewer connections. Immigrants who have spent longer in Sweden have larger networks, decreasing the effect of the average connection (Figure 4), potentially decreasing the total effectiveness of their networks relative to what we find assuming a size-invariant connection effect. Our conclusions are robust to allowing the connection effect to vary with network size, (Figure A.12), as our baseline estimates already average over effects for individuals with different network sizes and compositions.

rate of 0.9 per cent, from Hobijn and Şahin (2009).[34]

Figure 9: Contribution of networks to assimilation in unemployment rates



Notes: Contribution of networks to steady-state unemployment rates. Estimated steady-state unemployment rates are calculated by inferring job-arrival rates from observed job-finding of displaced workers in 1990–2016. The counterfactual steady-state unemployment rate is calculated by subtracting the total contribution of networks from the observed job-finding rates of displaced workers to back out a counterfactual job-arrival rate in the absence of networks.

Figure 9 shows the steady-state unemployment rates implied by these arrival rates. Newly arrived immigrants would face an unemployment rate of about 11.7 per cent, compared with 7.5 per cent for similar-aged natives, implying a gap of 4.2 percentage points.[35] As immigrants spend more years in Sweden, higher job-arrival rates lead them to close roughly 40–50 per cent of this initial unemployment gap.

To quantify how much of this convergence can be attributed to networks, we subtract the total effect of networks on re-employment shown in Figure 8 from the observed re-employment rates in Figure 2b. This allows us to back out a counterfactual job-arrival rate and steady-state unemployment in the absence of any accumulation of networks. We consider both the contribution of our narrow set of past coworkers (excluding large workplaces) and the contribution of past coworker networks when including large workplaces. The results imply that the accumulation of narrowly-defined networks reduces the unemployment gap by $(0.021 - 0.019)/4.1 = 6$ per cent after 10 years in Sweden, relative to the initial gap. More inclusively-defined networks reduce the unemployment gap by around 26 per cent after 10 years—more than half of the entire observed reduction in the steady-state employment gap.

[34] This figure is calculated using data from 1995–2004.

[35] Note that since the calculations are based on a sample of displaced individuals, they should not necessarily mirror the unemployment rates for immigrants and natives in general.

Taken together, these calculations suggest that the accumulation and growing effectiveness of immigrants' professional networks account for an important share of the long-run convergence in unemployment rates between immigrants and natives.

6 Conclusion

Networks play a central role in the functioning of labour markets. Immigrants, however, often arrive with few or no professional connections in the host country. This paper has examined how such networks develop after migration and how they influence immigrants' labour market integration, focusing specifically on connections formed through workplace interactions. In our setting, these networks arise from coworkers encountered in prior employment, rather than from deliberate networking behaviour in the colloquial sense.

Our first contribution is to document the dynamic evolution of professional networks after arrival. Using population-wide employer–employee data from Sweden, we construct detailed, time- and type-varying measures of individual networks. We show that immigrants' firm connections expand steadily with host-country tenure and include an increasing share of native contacts. This accumulation and diversification of professional relationships offer a new perspective on assimilation, complementing earlier work that emphasises human capital accumulation and firm sorting.

Our second contribution is to provide causal evidence on how different types of connections affect job finding, and how these effects evolve over time. Using a dyadic design and exploiting plant closures to identify exogenous job loss, we compare the re-employment outcomes of connected and unconnected displaced workers exposed to the same shock. Consistent with strong homophily in network use, we find that conational ties are far more valuable to immigrants than native ties, while native workers benefit most from native connections. Although other aspects of similarity (e.g. education and gender) and tie quality (e.g. recency and duration) also matter, shared origin remains the dominant factor shaping the usefulness of connections. When it comes to the mechanisms underlying these effects, patterns in post-re-employment earnings and turnover suggest that origin similarity between job seekers and their contacts leads to referrals, improving match quality.

Overall, our results point to persistent immigrant–native segregation in job finding and in the returns to professional contacts. Yet, we also document signs of convergence. Early in the assimilation process, improvements in job finding are driven mainly by the accumulation of high-productivity conational ties. Later, the value of immigrants' native connections rises, approaching that of native–native ties. In a decomposition, networks account for at least 20 per cent of the increase in re-employment probabilities during immigrants' first decade in Sweden, while the rising productivity of native ties accounts for about 10 per cent of the continued convergence observed thereafter. To our knowledge, this is the first evidence of a long-run assimilation process in the effectiveness of heterogeneous professional networks.

The study also has bearing on broader issues. Immigrant integration affects not only indi-

vidual outcomes but also the resilience and efficiency of host-country labour markets. Because immigrants are more exposed to both firm-level shocks and aggregate downturns, the insurance value of strong professional networks can be particularly important. From a policy perspective, our results suggest that integration initiatives that foster early contact with natives—especially in mixed workplaces—may help immigrants form the "bridging" ties that become increasingly valuable over time.. In the short run, co-national networks provide essential support for newcomers, but over time, professional connections with natives become critical for continued advancement. Programs that combine employment promotion with opportunities for intergroup contact—for example, through mentorship schemes, mobility incentives, or targeted placement services—could therefore enhance re-employment prospects and narrow immigrant–native gaps more rapidly.

Finally, while network effects are economically meaningful, they explain only part of immigrants' labour-market trajectories. Differences in search behaviour and job-application strategies (Brouard et al., 2024), together with language-related constraints (Carlsson et al., 2023; Foged et al., 2024), also contribute to persistent gaps. Evidence that stand-alone language training yields only modest integration effects (e.g. Foged et al., 2024) suggests that language proficiency enhances labour-market outcomes primarily when combined with other complementary forms of capital, including productive social networks. This points to potential complementarities between linguistic, social, and other host-country-specific resources. Future research should examine how language skills, broader human capital, and networks interact to shape immigrants' access to—and advancement within—host-country labour markets.

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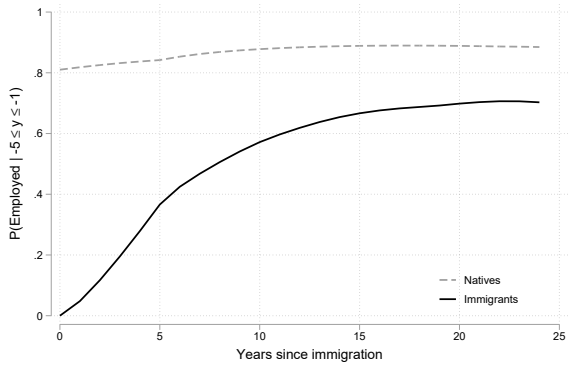
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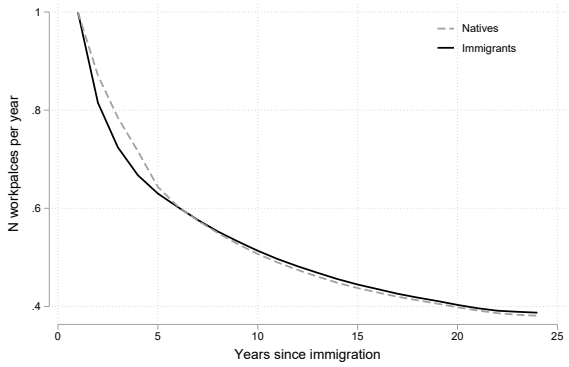
A Supplementary figures and tables

Figure A.1: Determinants of the number of recent coworkers

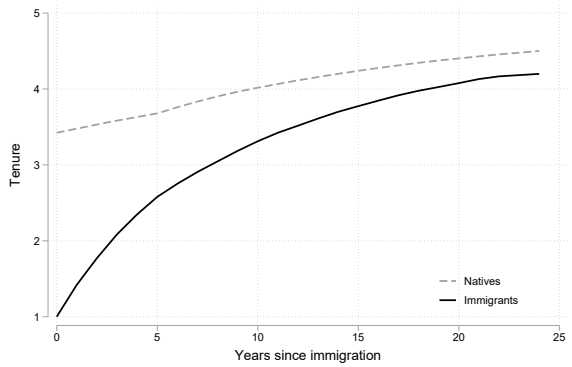
(a) Employment rate, $[y-5, y-1]$



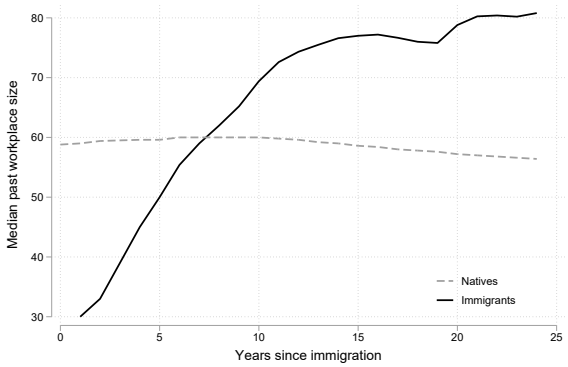
(b) Workplaces per year, $[y-5, y-1]$



(c) Tenure, y



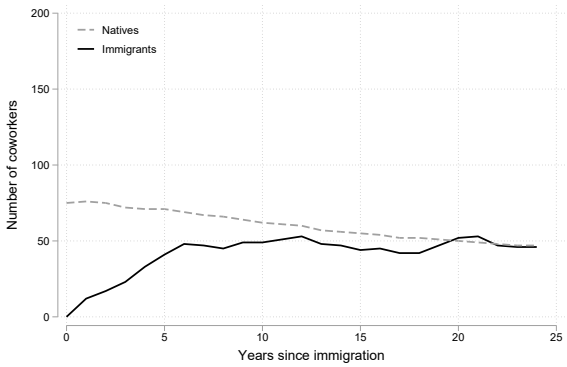
(d) Median workplace size, $[y-5, y-1]$



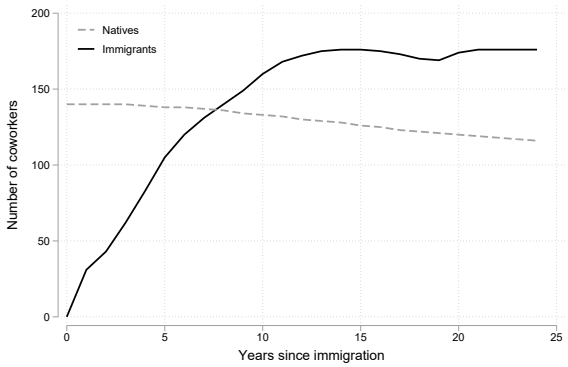
Notes: Average for immigrants by years since immigration and for same-aged natives, 1990–2016. Employment rate for all individuals not missing a workplace identifier during $[y-5, y-1]$. Workplaces per year, conditional on having been employed in $[y-5, y-1]$, further reweights natives by the distribution over number of years employed in $[y-5, y-1]$ among immigrants by years since immigration. Tenure is conditional on being employed in year y. Average workplace size is conditional on having been employed at some point in $[y-5, y-1]$.

Figure A.2: Number of recent coworkers by years since immigration

(a) Displaced individuals

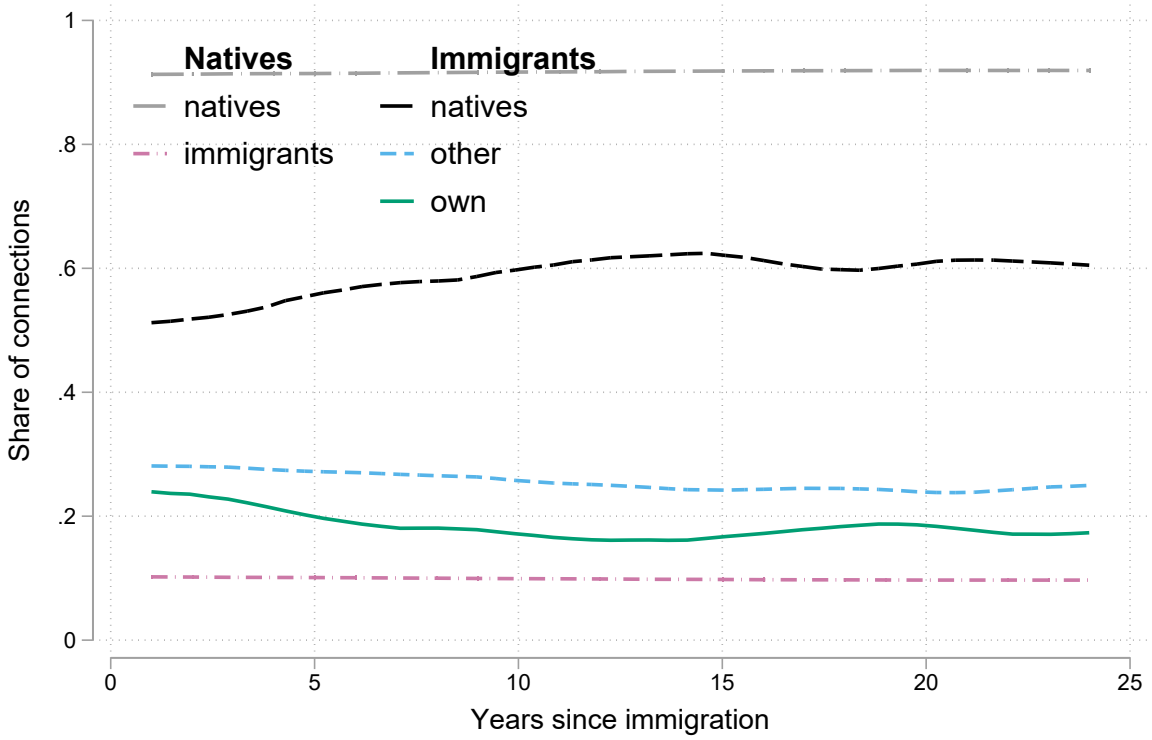


(b) All employed individuals



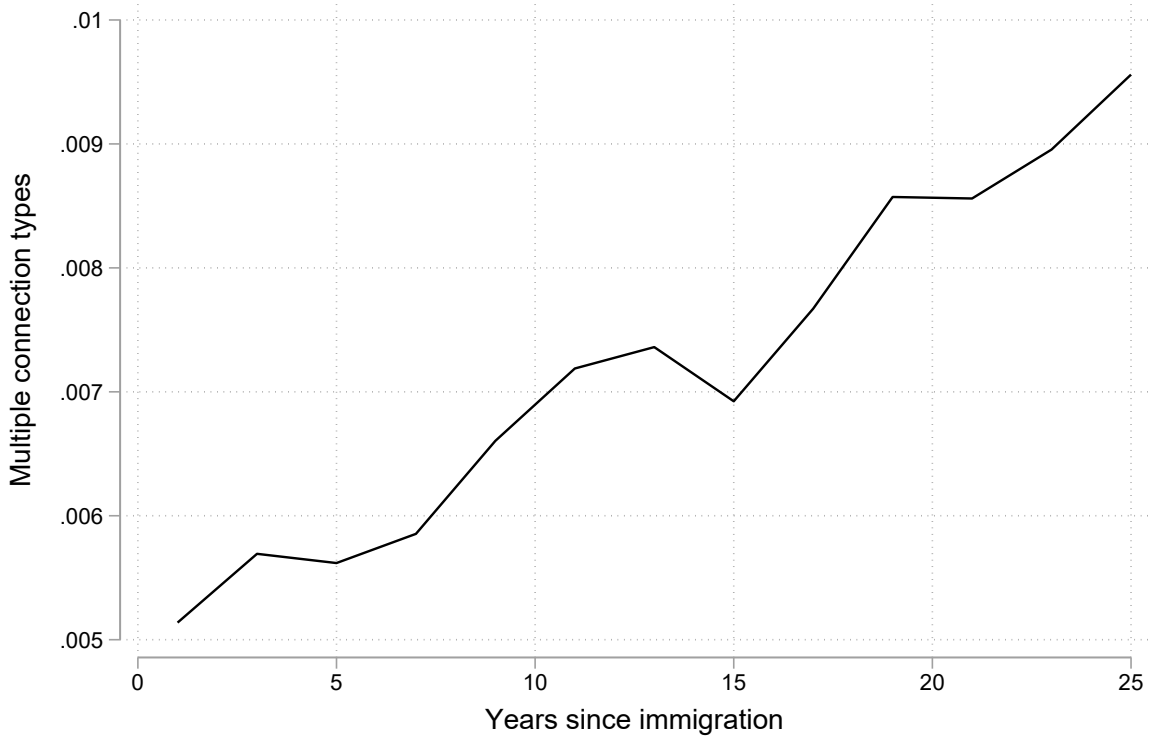
Notes: Median number of recent coworkers ($[y-5, y-1]$) for all individuals not missing a workplace identifier during that period, including individuals with no recent coworkers, for y in 1990–2016.

Figure A.3: Network composition over time in host country. Displaced workers.



Notes: Share of connections of different types at displacement by years since immigration. Kernel-weighted local averages are reported, calculated using the Epanechnikov kernel and a default bandwidth. The average number of native and immigrant connections for natives are weighted by the age distribution of immigrants, grouping age in five-year bins, conditional on years since immigration.

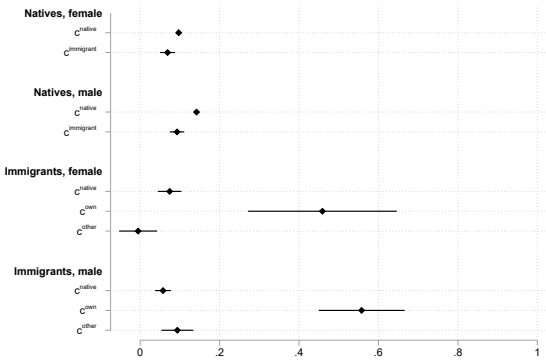
Figure A.4: Probability of having multiple types of connections



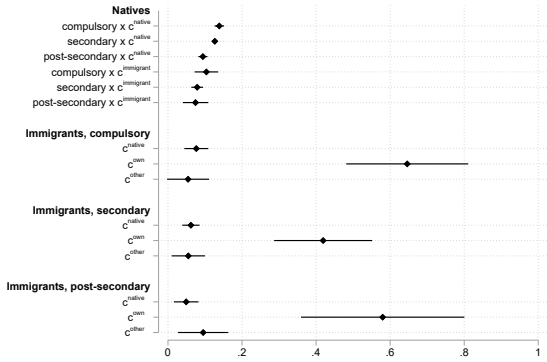
Notes: Probability of having multiple types of connections to a firm by years since immigration. Years since migration are grouped into two-year bins. An observation is a displaced worker by potential hiring firm to which they have at least one type of connection.

Figure A.5: Heterogeneity by other displaced worker characteristics

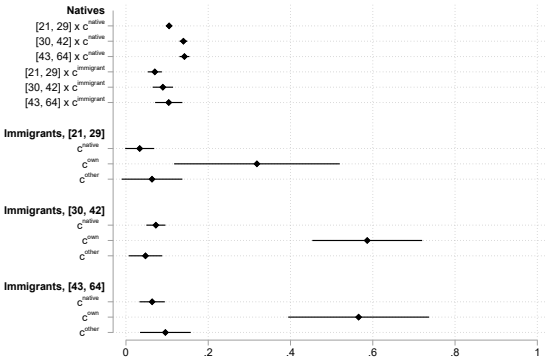
(a) Gender



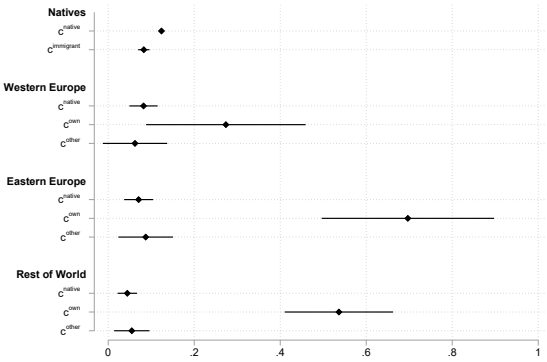
(b) Education



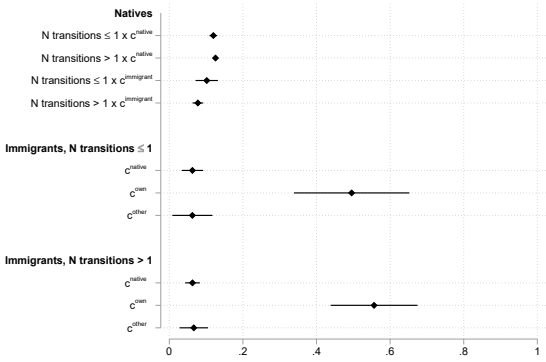
(c) Age



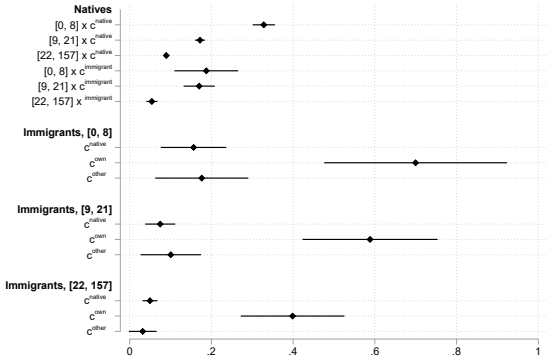
(d) Country of birth



(e) Past transitions



(f) N connections

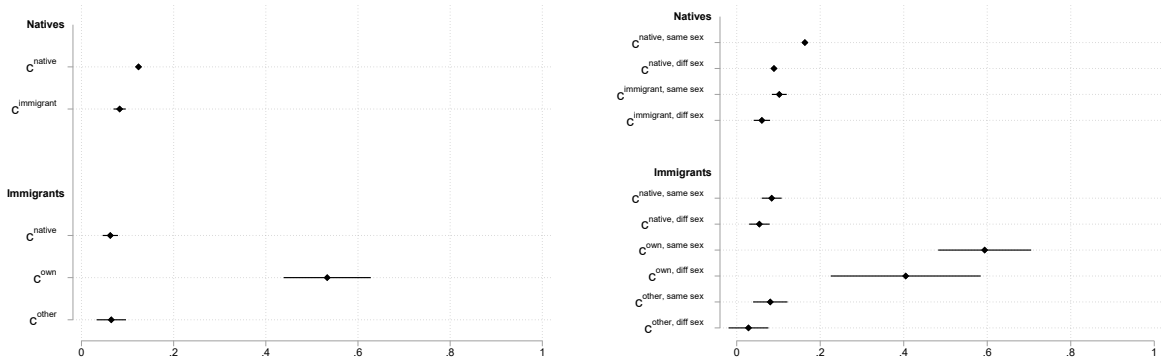


Notes: Coefficients estimated using Equation (1), allowing the effect of a connection on the hiring probability to vary by the nativity of the displaced and connecting workers as well as by other characteristics of the displaced worker. Western Europe includes the US and Canada, Eastern Europe includes Turkey and former members of the USSR. Standard errors are clustered by displaced individual i and 95 per cent confidence intervals are reported.

Figure A.6: Heterogeneity by homophily in other characteristics

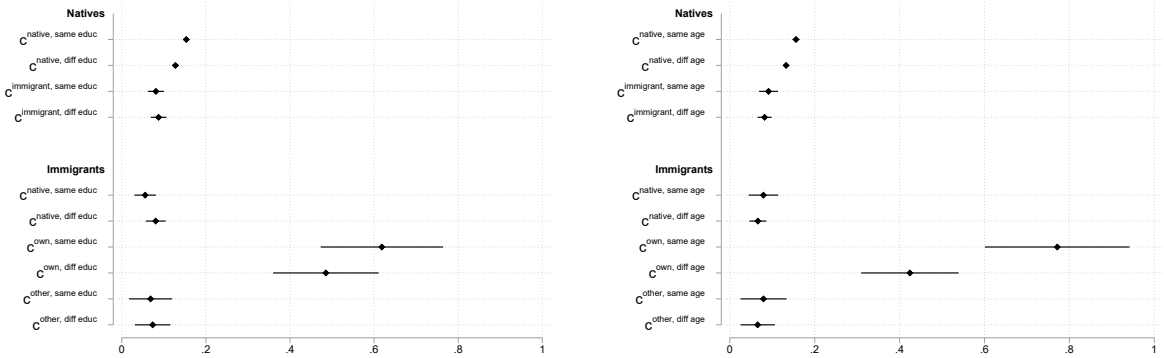
(a) No heterogeneity

(b) Gender homophily



(c) Education homophily

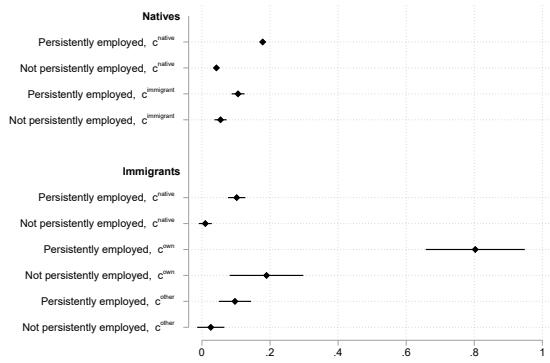
(d) Age homophily



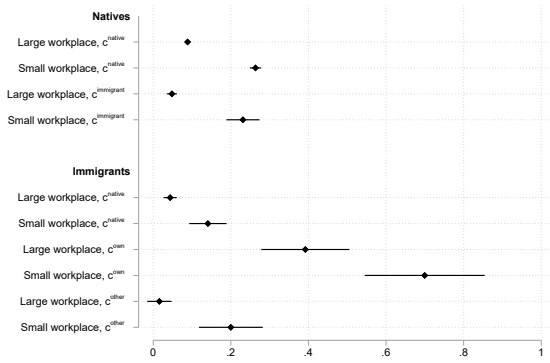
Notes: Coefficients estimated using Equation (1), allowing the effect of a connection on the hiring probability to vary by the nativity of the displaced and connecting workers as well as by homophily between both workers along other dimensions. Gender and education homophily refer to the displaced individual and at least one past coworker at the connected firm being the same gender or education (three categories). Age homophily refers to both individuals being within five years in age. Standard errors are clustered by displaced individual and 95 per cent confidence intervals are reported.

Figure A.7: Heterogeneity by predictors of tie quality

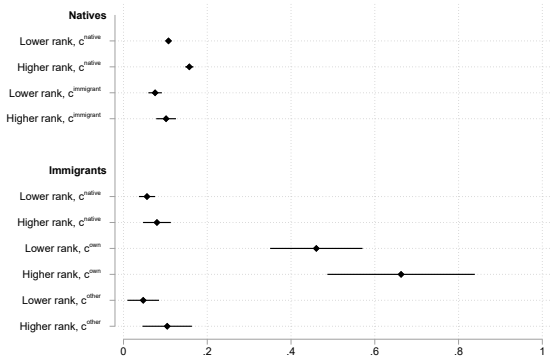
(a) Connection to k in t and $t+1$



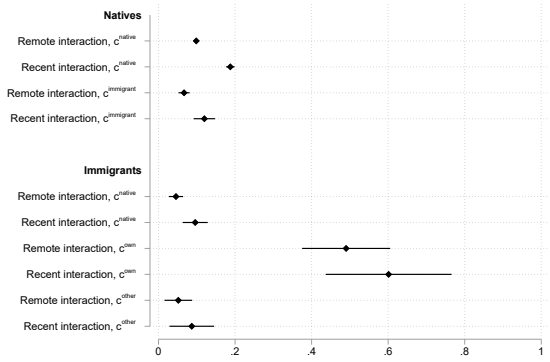
(b) Prior workplace size



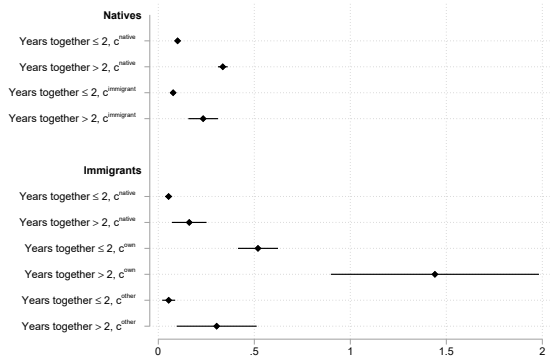
(c) Relative earnings



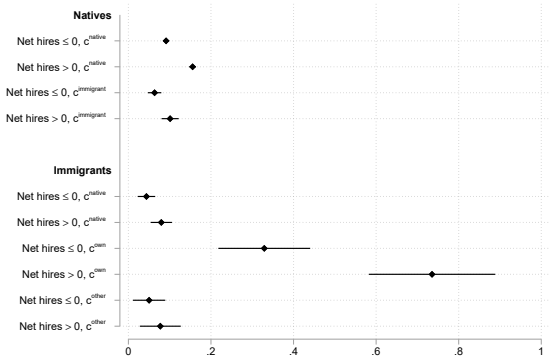
(d) Time since interaction



(e) Strong tie

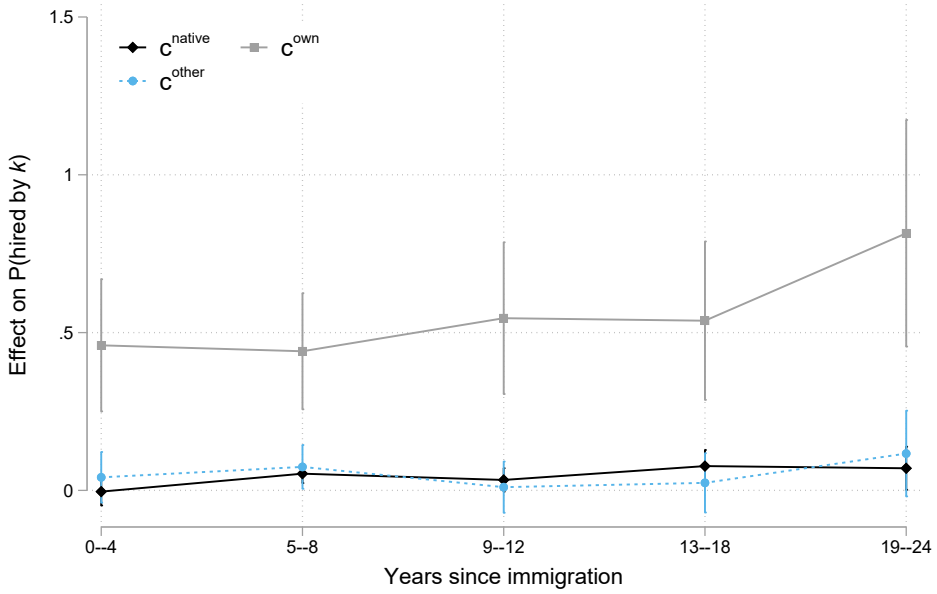


(f) Firm k expanded in t



Notes: Coefficients estimated using Equation (1), allowing the effect of a connection on the hiring probability to vary by the nativity of the displaced and connecting workers as well as by different measures of tie quality, measured as indicator variables. Small workplaces employ 20 workers or fewer. Higher ranked individuals were higher in the earnings ranking in the last prior workplace. Recent interactions occurred within the past two years. Strong ties are past coworkers with whom the displaced worker interacted for at least two years. Firm k expanded in if net hiring was positive in t. Standard errors are clustered by displaced individual and 95 per cent confidence intervals are reported.

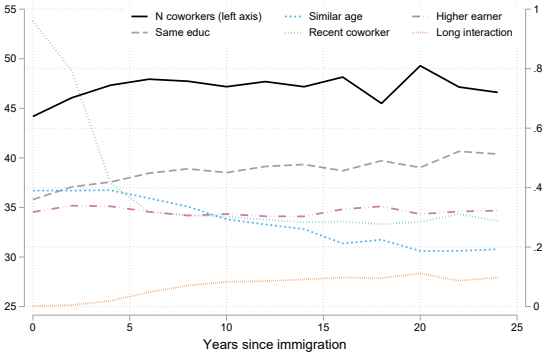
Figure A.8: Within-cohort effect of firm connections by years since immigration



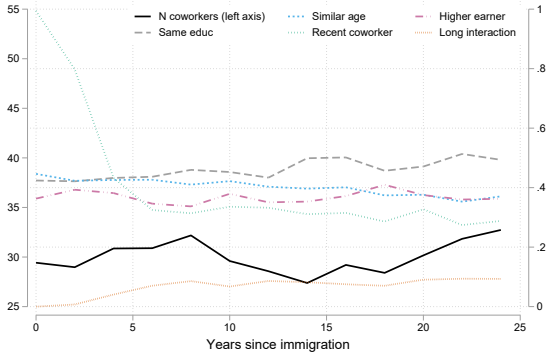
Notes: Heterogeneous effects of connections on the probability a displaced immigrant is hired by firm k, estimated by interacting the connectedness indicators for displaced immigrants with a set of indicators for years since migration at the time of displacement. Marginal effects of a connection by years in Sweden is shown. The firm-pair fixed effect is interacted with a set of cohort dummies. Standard errors are clustered by displaced individual i, 95 per cent confidence intervals are reported.

Figure A.9: Connection quality by years since migration

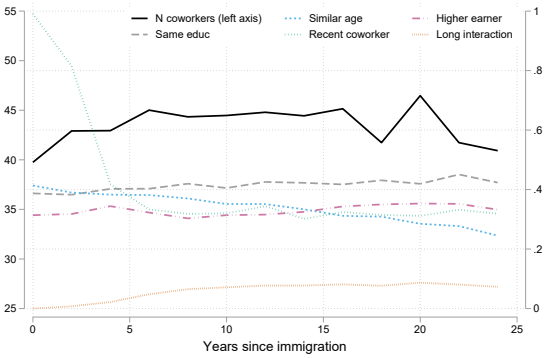
(a) Native connections



(b) Conational connections



(c) Other immigrant connections

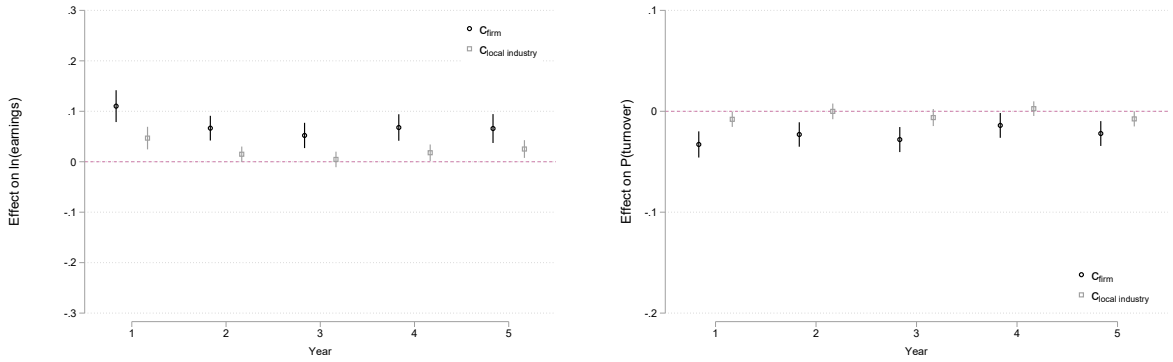


Notes: Average characteristics for a set of indicator variables that measure connection quality by years since immigration for displaced immigrants' native and conational connections. Higher ranked individuals were higher in the earnings ranking in the last prior workplace. Recent interactions occurred within the past two years. Same education and similar age are indicator variables equal to one if both displaced worker and at least one past coworker at a connected firm share a characteristic. Similar-age connections are within five years of age to the focal worker. Long interactions lasted two or more years. The number of coworkers in the workplace where the connection was formed is reported on the left axis.

Figure A.10: Labour market effects of local-industry connections

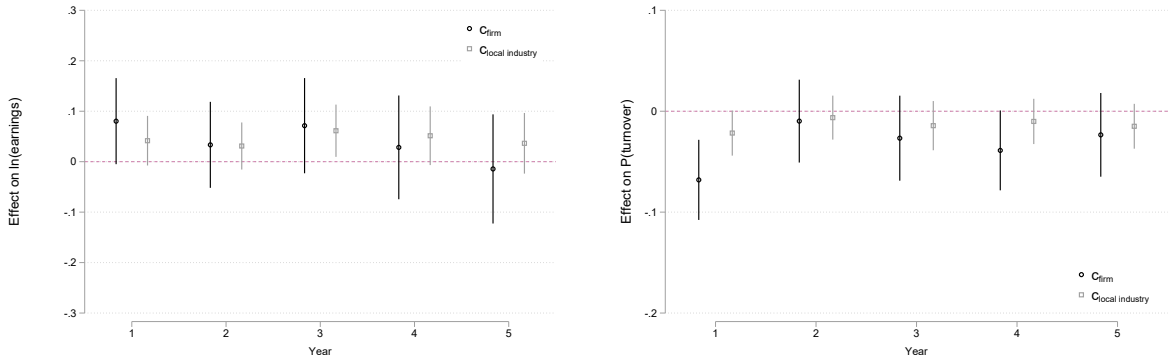
(a) Displaced natives, earnings

(b) Displaced natives, turnover



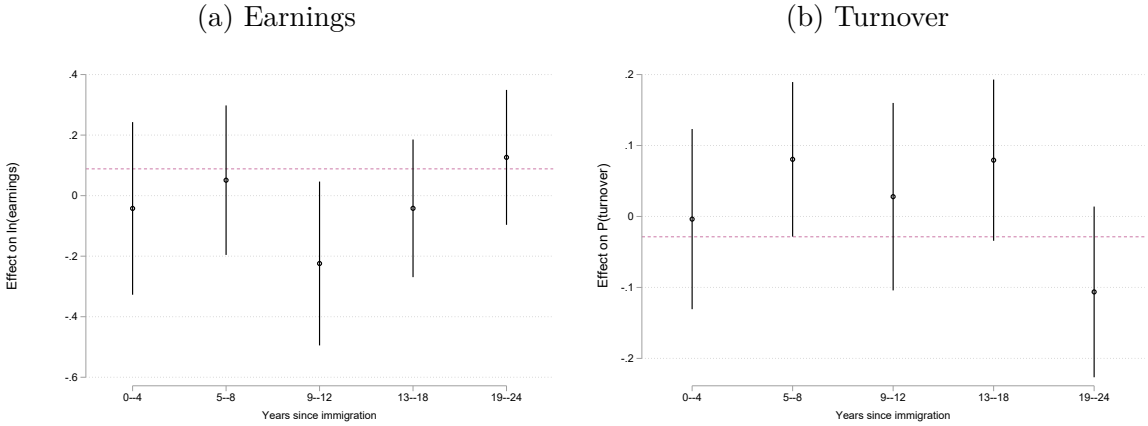
(c) Displaced immigrants, earnings

(d) Displaced immigrants, turnover



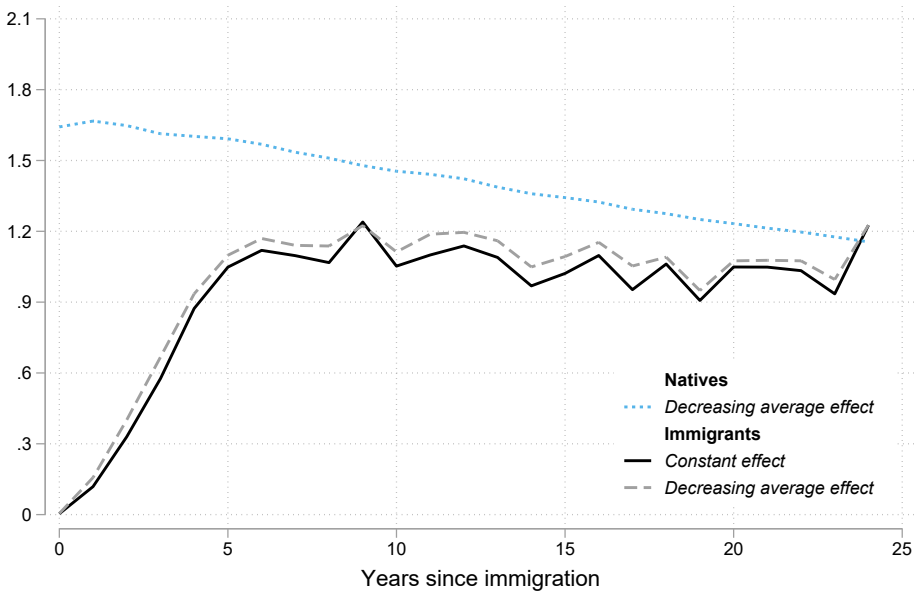
Notes: Coefficients estimated using Equation (2), allowing the effect of a connection at the hiring firm on subsequent outcomes to vary by the nativity of the displaced worker and whether the connection is at the hiring firm or at another firm in the same local industry. Standard errors are clustered by closing firm j and 95 per cent confidence intervals are reported.

Figure A.11: Effect of native connections for immigrants by years since immigration



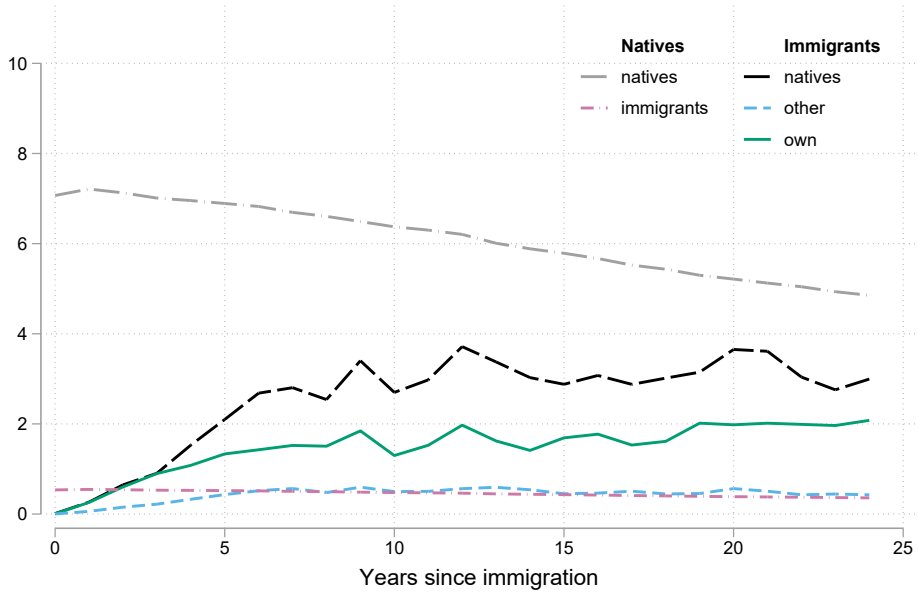
Notes: Coefficients estimated using Equation (2), allowing the effect of a connected hire on subsequent outcomes to vary by the nativity of the displaced and connecting workers as well as years since immigration. Estimates only reported for first post-displacement year, $t + 1$. Effects of a native connection for a hired native are reported as horizontal lines. Standard errors are clustered by closing firm j and 95 per cent confidence intervals are reported.

Figure A.12: Contribution of networks to assimilation: effects of connections varying with network size



Notes: Contribution of networks to re-employment in November of $t + 1$, defined according to Equation (C.2). Both constant (network size invariant) and varying-by-number-of-same-type estimates of the effect of a connection are considered. The contribution of natives' networks to their re-employment probability is calculated using the number of connections of different types for similar-aged natives for a given number of years since immigration.

Figure A.13: Contribution of networks to assimilation: including coworkers from large firms



Notes: Contribution of networks to re-employment in November of $t + 1$, defined according to Equation (C.2). Only time-invariant estimates of the effect of a connection are considered, from a specification that includes a sample of former coworkers from large workplaces. The contribution of natives' networks to their re-employment probability is calculated using the number of connections of different types for similar-aged natives for a given number of years since immigration.

Table A.1: Summary statistics for closing workplaces in t

	All firms	By workforce composition		
		Native-only	Mixed	Immigrant-only
N employees	10.3	6.83	13.9	5.78
	(22.3)	(4.42)	(30.8)	(3.18)
Immigrant share	0.18	0	0.29	1
	(0.26)		(0.21)	
Compulsory educ share	0.28	0.27	0.29	0.43
	(0.23)	(0.24)	(0.22)	(0.27)
Secondary educ share	0.59	0.61	0.57	0.39
	(0.24)	(0.25)	(0.23)	(0.25)
Tertiary educ share	0.12	0.11	0.13	0.18
	(0.19)	(0.19)	(0.19)	(0.21)
Industry				
Primary sector & utilities	0.028	0.035	0.023	0.017
	(0.17)	(0.18)	(0.15)	(0.13)
Manufacturing	0.13	0.12	0.13	0.084
	(0.33)	(0.33)	(0.34)	(0.28)
Construction	0.090	0.097	0.086	0.064
	(0.29)	(0.30)	(0.28)	(0.25)
Retail & repairs	0.22	0.28	0.17	0.099
	(0.41)	(0.45)	(0.38)	(0.30)
Hospitality	0.17	0.11	0.21	0.39
	(0.38)	(0.31)	(0.41)	(0.49)
TSC	0.067	0.064	0.066	0.13
	(0.25)	(0.24)	(0.25)	(0.33)
Corporate services	0.19	0.18	0.20	0.17
	(0.39)	(0.38)	(0.40)	(0.37)
Other services	0.086	0.088	0.088	0.041
	(0.28)	(0.28)	(0.28)	(0.20)
Observations	50212	23384	25083	1745

Note: Mean coefficients; standard deviations in parentheses. Firms are differentiated by workforce composition. TSC: Transport, storage and communications.

Table A.2: Population summary statistics

	Immigrants				Natives			
	Displ.	All	Nonem.	Emp.	Displ.	All	Nonem.	Emp.
Female	0.31	0.50	0.54	0.46	0.36	0.48	0.52	0.47
	(0.46)	(0.50)	(0.50)	(0.50)	(0.48)	(0.50)	(0.50)	(0.50)
Age	37.1	39.1	38.7	39.5	37.6	42.7	44.3	42.4
	(9.17)	(10.2)	(11.2)	(9.24)	(12.2)	(12.4)	(14.8)	(11.9)
Education								
Compulsory	0.35	0.36	0.48	0.26	0.22	0.19	0.35	0.16
	(0.48)	(0.48)	(0.50)	(0.44)	(0.42)	(0.39)	(0.48)	(0.37)
Secondary	0.41	0.35	0.29	0.40	0.65	0.55	0.52	0.56
	(0.49)	(0.48)	(0.45)	(0.49)	(0.48)	(0.50)	(0.50)	(0.50)
Post-Secondary	0.23	0.29	0.23	0.34	0.13	0.26	0.13	0.28
	(0.42)	(0.46)	(0.42)	(0.47)	(0.33)	(0.44)	(0.33)	(0.45)
N workplaces	1.36	0.90	0.35	1.37	1.97	1.52	0.92	1.63
	(1.05)	(0.96)	(0.69)	(0.91)	(1.02)	(0.91)	(0.98)	(0.85)
Years employed	2.50	2.09	0.61	3.33	4.20	4.23	1.85	4.65
	(1.89)	(2.12)	(1.24)	(1.88)	(1.28)	(1.50)	(1.90)	(0.90)
Tenure in firm	1.99	3.10	.	3.10	2.67	4.03	.	4.03
	(1.49)	(2.00)	(.)	(2.00)	(1.93)	(2.05)	(.)	(2.05)
Emigrated by $y+1$	0.024	0.028	0.044	0.014	0.0022	0.0014	0.0038	0.00094
	(0.15)	(0.16)	(0.20)	(0.12)	(0.047)	(0.037)	(0.061)	(0.031)
N past coworkers	229.8	826.3	508.8	904.8	287.8	747.1	510.0	772.4
	(882.6)	(2162.4)	(1550.6)	(2282.0)	(927.6)	(1900.5)	(1426.7)	(1942.5)
N (000s)	44	12,129	5,587	6,542	278	98,351	15,078	83,273

Note: Displaced immigrant and native (column 1 and 5) and population (columns 2–4 and 6–8) means, 1990–2016; standard deviations in parentheses. Individuals in columns 2–4 and 6–8 are weighted according to the distribution across calendar years of workplace closures in the analysis sample. "Displ." refers to displaced individuals, "Nonem." refers to individuals who are not employed. Sex, education categories, and emigration by the end of $y+1$ are measured as indicator variables. N workplaces, Years employed, and Tenure in firm are all measured over $[y-5, y-1]$. Compulsory education includes individuals with missing education information.).

Table A.3: Determinants of displaced workers' connections

		Years since immigration				
	Native	0–4	5–9	10–14	15–19	20–24
Prior employment	0.98	0.58	0.90	0.94	0.95	0.96
	(0.15)	(0.49)	(0.30)	(0.24)	(0.22)	(0.20)
N past workplaces > 0	0.76	0.57	0.78	0.76	0.74	0.73
	(0.43)	(0.49)	(0.42)	(0.42)	(0.44)	(0.45)
N past coworkers, $[t-5, t-1]$	245.0	53.3	206.2	243.4	227.8	246.1
	(908.9)	(413.8)	(896.2)	(965.3)	(968.9)	(1021.1)
N uncensored past coworkers, $[t-5, t-1]$	30.6	7.81	23.4	24.3	22.4	22.7
	(41.7)	(20.3)	(36.7)	(37.4)	(36.0)	(37.5)
Share censored coworkers	0.30	0.20	0.27	0.29	0.27	0.29
	(0.43)	(0.39)	(0.42)	(0.43)	(0.43)	(0.43)
N uncensored natives, $[t-5, t-1]$	27.1	4.41	14.3	16.0	14.5	14.7
	(37.0)	(13.8)	(26.2)	(28.1)	(26.5)	(27.4)
N uncensored conationals, $[t-5, t-1]$		1.21	2.78	2.32	2.41	2.71
		(4.77)	(7.09)	(5.80)	(6.24)	(7.20)
N uncensored others, $[t-5, t-1]$	3.46	2.20	6.35	5.95	5.51	5.23
	(7.23)	(7.27)	(13.2)	(12.5)	(12.1)	(12.5)
Conational emigration rate		0.068	0.075	0.079	0.092	0.12
		(0.18)	(0.18)	(0.19)	(0.20)	(0.22)
Other migrant emigration rate		0.079	0.089	0.092	0.093	0.094
		(0.17)	(0.16)	(0.16)	(0.16)	(0.17)
N	278,993	18,064	10,418	6,669	5,157	4,383

Note: Summary statistics of the main factors that determine the number of connections a displaced individual has in the year of displacement. Past job transition is reported conditional on having a prior employment in the network building phase. Past coworkers are from workplaces other than the closing workplace. Coworkers are censored if the past workplace employed more than 100 workers; share censored coworkers is conditional on having past coworkers. For natives, "N uncensored others" refers to immigrant past coworkers of all birth places. Emigration rates by past coworker type are reported conditional on having a past coworker of the given type. Emigration rates are the share of coworkers from the last year of employment in the prior workplace who have emigrated by the end of the last pre-displacement year, t, with no further restrictions. Sample standard deviations are reported in parentheses.

Table A.4: Effects of connections on firm hiring

	(1) Hired by k	(2) Hired by k	(3) Hired by k	(4) Hired by k	(5) Hired by k	(6) Hired by k	(7) Hired by k
Natives							
c^{native}	0.12	0.17	0.12	0.11	0.12	0.12	0.12
	(0.003)	(0.005)	(0.002)	(0.002)	(0.003)	(0.003)	(0.003)
$c^{immigrant}$	0.083	0.12	0.081	0.055	0.080	0.091	0.083
	(0.007)	(0.01)	(0.006)	(0.006)	(0.007)	(0.008)	(0.008)
Immigrants							
c^{native}	0.063	0.083	0.064	0.065	0.067	0.067	0.063
	(0.008)	(0.02)	(0.008)	(0.007)	(0.009)	(0.009)	(0.010)
c^{own}	0.53	0.61	0.50	0.50	0.51	0.51	0.53
	(0.05)	(0.08)	(0.04)	(0.04)	(0.05)	(0.06)	(0.05)
c^{other}	0.065	0.11	0.077	0.039	0.058	0.060	0.065
	(0.02)	(0.03)	(0.02)	(0.01)	(0.02)	(0.02)	(0.02)
$\hat{\mathrm{E}}[Y(0)\|c^{any}=0]$	0.023	0.023	0.022	0.012	0.023	0.020	0.023
N (000s)	68,601	66,217	87,452	165,151	40,440	61,183	68,601

Note: Column 1 reports our basic specification. Column 2 restricts the sample to only include connections through coworkers from the last workplace prior to the closing establishment. Column 3 includes individuals who previously worked at a workplace with a missing identifier and assumes that they formed no connections in that workplace. Column 4 returns to the main sample and randomly samples 100 coworkers in larger workplaces to define connections formed in that workplace. Column 5 restricts the sample to individuals with at least one connection. Column 6 restricts the sample to closing firms that employed both immigrants and natives. Column 7 repeats the basic specification, clustering standard errors by hiring firm k. Standard errors are clustered by displaced individual i in all other specifications.

Table A.5: Effect of a placebo connection

	(1) $k_h ire$	(2) Hired by k	(3) Hired by k	(4) Hired by k
c^{native} × Native	0.12***	0.039***	0.034***	0.029***
	(0.0029)	(0.0012)	(0.0012)	(0.0013)
$c^{immigrant}$ × Native	0.084***	0.028***	0.024***	0.024***
	(0.0078)	(0.0031)	(0.0031)	(0.0036)
c^{native} × Immigrant	0.064***	0.027***	0.023***	0.023***
	(0.0094)	(0.0038)	(0.0040)	(0.0047)
c^{own} × Immigrant	0.56***	0.076***	0.068***	0.054**
	(0.054)	(0.015)	(0.015)	(0.018)
c^{other} × Immigrant	0.071***	0.035***	0.031***	0.033***
	(0.018)	(0.0066)	(0.0069)	(0.0091)
Controls	Yes	No	Yes	Yes
Fixed Effects	jk	jk	jk	$jk \times IM$
$\hat{\mathrm{E}}[Y(0)\|c^{any}=0]$	0.021	0.00043	0.00043	0.00039
N	63080712	47743821	47279293	45736990

Note: Column (1) reports the baseline estimates for actual connections. Columns (2)–(4) report variations of estimates for placebo connections. See main text for description of the procedure. IM is an indicator for having previously worked in the industry and municipality of firm k in a displaced worker's previous job. Standard errors clustered by displaced individual, 95 per cent confidence intervals are reported.

Table A.6: Summary statistics, local industry dyads

		Years since immigration				
	Native	0–4	5–9	10–14	15–19	20–24
N workplaces	389.2 [91]	553.1 [123]	482.5 [111]	436.4 [100]	409.3 [91]	367.4 [78]
N connected workplaces						
All connections	0.1	0.03	0.09	0.09	0.07	0.07
Natives	0.10	0.02	0.06	0.07	0.05	0.05
Immigrants	0.007	0.01	0.03	0.02	0.02	0.02
Conationals		0.004	0.009	0.007	0.006	0.006
Other immigrants	0.007	0.008	0.02	0.02	0.01	0.01
Multiple connected workplaces	0.09	0.03	0.08	0.08	0.07	0.06
N (000s)	25,237	1,063	762	511	423	383

Note: We report mean [median] values of different variables defined for each displaced worker by potential hiring local industry included in our sample. Multiple connected workplaces is an indicator equal to one if a displaced individual is connected to more than one workplace in the local industry.

Table A.7: Effects of connections on industry hiring

	(1) Hired in k	(2) Hired in k^C	(3) Hired in k^{UC}	(4) Hired in k	(5) Hired by k
c^{native} × Native	0.31***	0.19***	0.12***	0.14***	0.11***
	(0.0062)	(0.0027)	(0.0057)	(0.0056)	(0.0026)
$c^{immigrant}$ × Native	0.11***	0.087***	0.028	0.13***	0.056***
	(0.019)	(0.0073)	(0.018)	(0.018)	(0.0071)
c^{native} × Immigrant	0.14***	0.11***	0.025	0.039	0.052***
	(0.023)	(0.0093)	(0.021)	(0.021)	(0.0089)
c^{own} × Immigrant	0.63***	0.69***	-0.057	0.47***	0.48***
	(0.099)	(0.058)	(0.085)	(0.093)	(0.055)
c^{other} × Immigrant	0.055	0.098***	-0.043	0.059	0.038*
	(0.047)	(0.017)	(0.044)	(0.044)	(0.017)
Controls	Yes	Yes	Yes	Yes	Yes
Fixed Effects	jk	jk	jk	jk	jk
$\hat{\mathrm{E}}[Y(0)\|c^{any}=0]$	0.16	0	0.16	0.16	0.023
N	28381771	28381771	28381771	27988723	48764765

Note: The dependent variable in columns 1–4 is an indicator for being hired in industry k, in column 5 it is an indicator for being hired by firm k. Columns 2 and 3 decompose the effect on industry hiring (column 1) into hires at the subset of firms where a past coworker is employed (column 2) and the remainder of firms in the industry (column 3). The sample of column 4 is restricted to industries k that a displaced worker is connected to by at most one past coworker. The dependent variable is scaled by 100. Controls are share native, conational, or same education as i, indicators for i having worked in same industry or municipality as k or (in the firm regression) sharing birth country with k's manager, all interacted with an indicator for i's nativity, and categories for region of birth. Standard errors clustered by displaced individual i are reported in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A.8: Firm connections at hiring and post-displacement outcomes

	(1) ln (earnings)	(2) turnover	(3) ln (earnings)	(4) turnover
$t+1 \times c_{firm}$	0.15***	-0.032***	0.096***	-0.030***
	(0.012)	(0.0055)	(0.015)	(0.0065)
$t+2 \times c_{firm}$	0.088***	-0.016**	0.060***	-0.022***
	(0.012)	(0.0056)	(0.012)	(0.0061)
$t+3 \times c_{firm}$	0.067***	-0.018**	0.048***	-0.026***
	(0.013)	(0.0057)	(0.013)	(0.0061)
$t+4 \times c_{firm}$	0.074***	-0.0092	0.060***	-0.014*
	(0.013)	(0.0058)	(0.013)	(0.0061)
$t+5 \times c_{firm}$	0.065***	-0.015*	0.056***	-0.020**
	(0.013)	(0.0059)	(0.014)	(0.0061)
$t+1\times$ Immigrant $\times c_{firm}$	-0.013	-0.022	-0.044	-0.032
	(0.039)	(0.019)	(0.044)	(0.020)
$t+2\times$ Immigrant $\times c_{firm}$	-0.038	0.0087	-0.037	0.014
	(0.043)	(0.019)	(0.044)	(0.021)
$t+3\times$ Immigrant $\times c_{firm}$	0.012	0.0074	0.0066	0.0068
	(0.046)	(0.020)	(0.048)	(0.021)
$t+4\times$ Immigrant $\times c_{firm}$	-0.037	-0.018	-0.045	-0.021
	(0.048)	(0.021)	(0.051)	(0.021)
$t+5\times$ Immigrant $\times c_{firm}$	-0.059	0.0050	-0.078	0.0021
	(0.050)	(0.021)	(0.053)	(0.021)
controls	No	No	Yes	Yes
Observations	744341	824628	736712	815669

Note: Included controls are indicators for having previously worked in the hiring firm or the hiring firm's industry, an indicator for having a conational manager at the hiring firm, the share of coworkers at the hiring firm with the same education as the displaced worker, the share of natives, the share of conationals and a quadratic in pre-displacement earnings, all interacted with dummies for the observation year post-hire, as well as gender-specific dummies for level of education and birth region as well as a quadratic in age, and nativity-specific terms for tenure at the time of displacement, the number of years worked of the five years prior to displacement, and the n umber of distinct workplaces in the five years prior to displacement. Standard errors are clustered by closing firm, j, and 95 per cent confidence intervals are reported.

B Identification with network data

Here we provide formal details for the "no-spillovers" assumption for empirical identification. We first consider a heterogeneous effects setting before relating the discussion to the constant-effects model we use in the text. For discussion purposes, we abstract from the multiple connection types setup used in the applications. Consider the set of I workers displaced by the closure of workplace j. Suppose there are K potential hiring workplaces. Let Y_{ik} denote an indicator for whether displaced worker i is hired by firm k, let c_{ik} be an indicator for whether worker i is connected to firm k, and let κ be a $K \times I$ dimensional matrix whose $(k,i)^{\text{th}}$ entry is equal to c_{ik}, i.e., is one if displaced worker i is connected to firm k and zero otherwise. For each displaced worker-hiring firm pair ik , there are 2^{KI} potential outcomes for Y_{ik} to consider, denoted $Y_{ik}(\kappa)$—one for each possible realisation of κ. We will denote $Y_{ik}(\kappa) = Y_{ik}(c_{ik}, \kappa_{-ki})$ where κ_{-ki} is the matrix κ with its $(k,i)^{\text{th}}$ entry set to zero.

For a given realisation of κ_{-ki}, we define the *conditional* causal effect of a connection on the probability of being hired as follows:

$$\gamma_{ik,-\kappa} = Y_{ik}(c_{ik} = 1, \kappa_{-ki}) - Y_{ik}(c_{ik} = 0, \kappa_{-ki}). \tag{B.1}$$

This effect is *conditional* in the sense that the causal effect of a connection to firm k on the probability that i is hired by the firm is a function of the topology of the entire network. We will impose two restrictions on the potential outcomes: (A1) the potential outcome $Y_{ik}(c_{ik}, \kappa_{-ki})$ does not depend on the entries in the columns of κ other than column i; and (A2) the potential outcome $Y_{ik}(c_{ik}, \kappa_{-ki})$ does not depend on the entries in any row other than row k. With these two "no spillover" assumptions, the *unconditional* causal effect of a connection on the probability of being hired is well-defined:

$$\gamma_{ik} = Y_{ik}(c_{ik} = 1) - Y_{ik}(c_{ik} = 0) \tag{B.2}$$

These two assumptions are a version of the standard Stable Unit Treatment Value Assumption (SUTVA) in the context of network data. Intuitively, assumption A1 states that i's chances of being hired by firm k do not depend on which firms their co-displaced workers are connected to; in particular, they do not depend on whether a co-displaced worker is connected to firm k or not. Assumption A2 is similar in spirit and states, intuitively, that i's chances of being hired by firm k do not depend on which other firms i (and i's co-displaced) are connected to, only whether i is connected to firm k. To obtain the constant-effects model of the text, we further need to assume that the causal effect γ_{ik} is also independent of the non-network characteristics of both the potential hiring firm, k, and of the displaced worker, i.

Neither A1 nor A2 are formally testable, as they depend on unobserved potential outcomes. Nevertheless, we do provide some evidence that these assumptions are likely to hold, at least approximately. In Figure B.1 we plot sample analogues of $\mathrm{E}[Y_{ik}(0)|c_{ik} = 0]$ and $\mathrm{E}[Y_{ik}(1)|c_{ik} = 1]$ conditional on different summaries of the matrix κ. Observed potential outcomes are residualised

within firm dyad jk, consistent with our empirical design; the sample mean of the hiring outcome is added to the residualised variables.

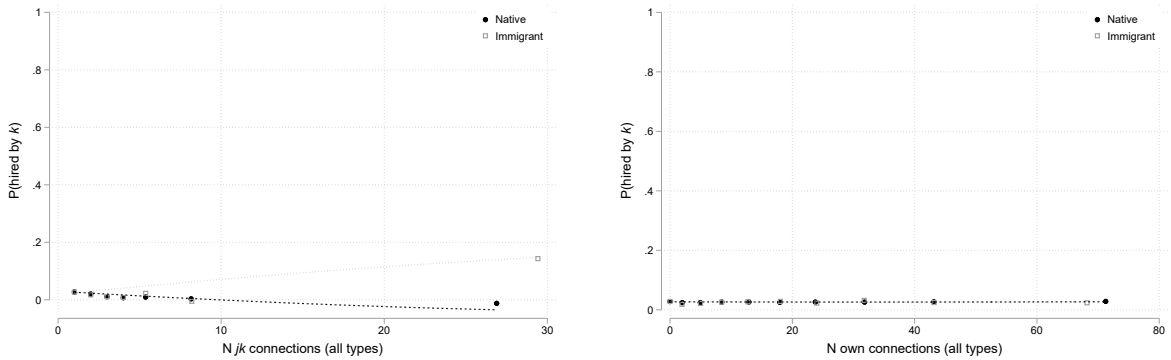
In the first column, we plot how the residualised potential outcomes vary as a function of the total number of co-displaced workers who are also connected to firm k (the x-axis shows the sum of the entries in row k of matrix κ). In the top row, we report potential outcomes for market (unconnected) hires, $Y(0)$, and find that these are independent of the number of co-displaced workers connected to hiring firm k. In rows two and three we report potential outcomes for connected hires, $Y(1)$, differentiating between connections through natives (row 2) and immigrants (row 3), as a function of the total number of co-displaced connected to workplace k. We again find little association, although there is some evidence of conational connections having a larger effect when more displaced coworkers are connected to firm k. We conclude that Assumption A1 is reasonable.

In the second column of Figure B.1, we plot how the residualised potential outcomes vary as a function of the total number of firms the displaced individual i is connected to (the x-axis shows the sum of entries in column i of the matrix κ). In the top row, we report potential outcomes for market hires, $Y(0)$, and find that these are independent of the total number of connections a worker has. In rows two and three we again report potential outcomes for connected hires, $Y(1)$, differentiating between connections through natives (row 2) and immigrants (row 3), as a function of the total number of connections. We find that a single connection is less likely to lead to a hire, the more connections a displaced individual has, violating assumption A2. This finding holds across both displaced worker and connection types.

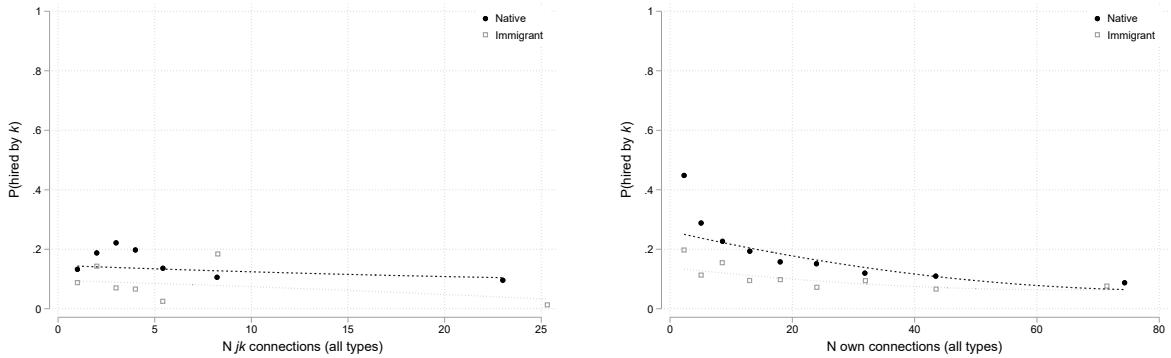
Since we do find evidence that Assumption A2 may not hold, in the main text we confirm the robustness of our results to both (i) comparing only displaced individuals with similarly-sized networks; and (ii) allowing the effect of connections to vary as a function of either the total number of connections, or the number of connections of the same type. In the constant-effects framework outlined in the text, where $Y_{ijk}(0) = \alpha_{jk} + X'_{ik}\beta + \epsilon_{ijk}$, we achieve (i) by interacting the fixed effect with categories for the size of the displaced worker's network. We achieve (ii) by interacting the connectedness indicators c_{ik} with measures of either the total number of connections, or the number of same-type connections.

Figure B.1: Potential hiring outcomes and network topology

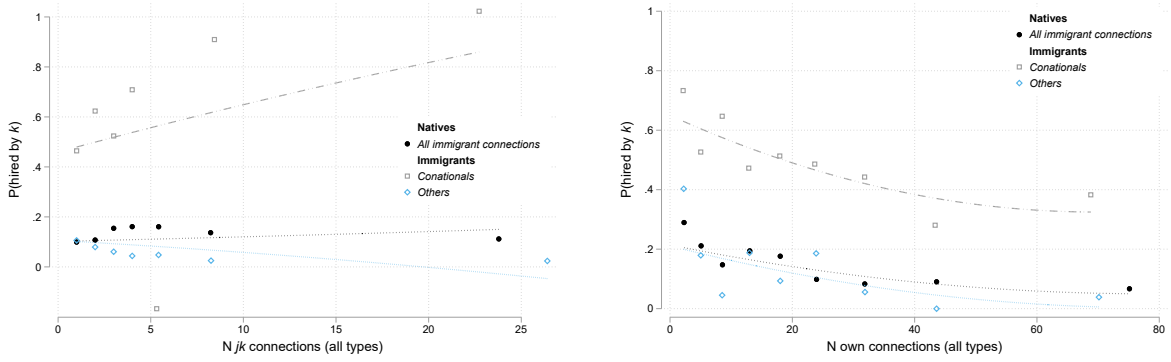
(a) $Y(k_i = 0)$: No connection to workplace k



(b) $Y(k_i = 1)$: Native connection to workplace k



(c) $Y(k_i = 1)$: Immigrant connection to workplace k



Notes: Binned scatter plot of an indicator for whether a displaced individual is hired by potential hiring firm k. The hiring indicator is residualised within firm dyads jk, the sample mean of each hiring variable is added to the residualised variables. The lines in each plot show fitted values from a quadratic fit.

C The total effect of the network

Here we show how the total effect of the network on the employment probability can be recovered from our dyadic estimates. For clarity of exposition, we first abstract from different connection types, time-varying effects, or effects that vary with network size. Let c_k denote an indicator for whether a displaced worker has a connection to firm k. Since a displaced worker can only be hired by one hiring firm k, the probability that a displaced worker is employed in November of $t+1$ can be decomposed using the law of total probability as follows:

$$\begin{aligned} P(\text{employed}) &= \sum_{k \in K} P(\text{hired by } k) \\ &= \sum_{k \in K} P(c_k = 1) P(\text{hired by } k | c_k = 1) + P(c_k = 0) P(\text{hired by } k | c_k = 0) \\ &= \sum_{k \in K} P(c_k = 1)\gamma + P(\text{hired by } k | c_k = 0) \\ &= \gamma \times \sum_{k \in K} c_k + \sum_{k \in K} P(\text{hired by } k | c_k = 0), \end{aligned} \tag{C.1}$$

where the third equation follows from the assumption of a constant effects model, where connections raise the probability of being hired by a firm by some constant $\gamma = P(\text{hired by } k|c_k = 1) - P(\text{hired by } k|c_k = 0)$. $\sum_{k \in K} c_k$ is simply the displaced worker's total number of connections.

To illustrate how to expand the decomposition framework to accommodate multiple connection types, consider two types of connections, immigrant and native, denoted c_k^I and c_k^N. The decomposition now proceeds as follows:

$$\begin{aligned} P(\text{employed}) &= \sum_{k \in K} P(\text{hired by } k) \\ &= \sum_{k \in K} \Big[P(c_k^N = 1, c_k^I = 1) P(\text{hired by } k | c_k^N = 1, c_k^I = 1) + \\ &\qquad P(c_k^N = 1, c_k^I = 0) P(\text{hired by } k | c_k^N = 1, c_k^I = 0) + \\ &\qquad P(c_k^N = 0, c_k^I = 1) P(\text{hired by } k | c_k^N = 0, c_k^I = 1) + \\ &\qquad P(c_k^N = 0, c_k^I = 0) P(\text{hired by } k | c_k^N = 0, c_k^I = 0) \Big] \\ &= \sum_{k \in K} \Big[P(c_k^N = 1, c_k^N = 1)(\gamma^N + \gamma^I) + \\ &\qquad P(c_k^N = 1, c_k^N =) \gamma^N + P(c_k^N = 0, c_k^N = 1)\gamma^I + \\ &\qquad P(\text{hired by } k | c_k^N = 0, c_k^I = 0) \Big] \\ &= \gamma^N \sum_{k \in K} c_k^N + \gamma^I \sum_{k \in K} c_k^I + \sum_{k \in K} P(\text{hired by } k | c_k = 0), \end{aligned} \tag{C.2}$$

The second equation again follows form the law of total probability, while the third equation follows from the constant effects model and an assumption that the effect of multiple different connection types to a firm on the hiring probability is additively separable into the effect of each

connection type.